Controller-RK3562 CONTROLLER TOP

ANTI-FALL-IR-SENSORs.SchDoc

ANTI-FALL IR

ANTI-FALL-LEFT-UP-ADC
ANTI-FALL-LEFT-DOWN-ADC
ANTI-FALL-RIGHT-UP-ADC
ANTI-FALL-RIGHT-DOWN-ADC

SIDE-PROXIMITY-IR-SENSOR .SchDoc

SIDE PROXI SENSOR

SIDE-PROXI-LEFT
TOUCHL
SIDE-PROXI-RIGHT
TOUCHR

DOCKING IR-BUMPER-SENSOR.SchDoc

DOC-IR BUMBER SW

DOC-IR-SENS1
DOC-IR-SENS2
BUMPER-SW1
BUMPER-SW2

WATER-PUMP .SchDoc

WATER PUMP

WATER-PUMPU-CTRL
WATER-PUMP-SENSE-ADC

MAIN-BRUSH-MOTOR .SchDoc

MAIN BRUSH MOTOR

MAIN-BRUSH
MAIN-BRUSH-F-ADC

WHEEL-MOTORs .SchDoc

WHEEL MOTORs

WHEEL-M-RIGHT-IN2
WHEEL-M-RIGHT-IN1
WHEEL-M-RIGHT-ENCODE-A
WHEEL-M-EN-V
WHEEL-M-LEFT-IN2
WHEEL-M-LEFT-IN1
WHEEL-M-LEFT-ENCODE-A
WHEEL-M-R-F-ADC
WHEEL-M-L-F-ADC

STM32G070RBT6.SchDoc

MCU-STM32

ANTI-FALL-RIGHT-DOWN-ADC
ANTI-FALL-RIGHT-UP-ADC
ANTI-FALL-LEFT-DOWN-ADC
ANTI-FALL-LEFT-UP-ADC
SIDE-PROXI-LEFT TOUCHL
SIDE-PROXI-RIGHT TOUCHR
DOC-IR-SENS1
DOC-IR-SENS2
BUMPER-SW1
BUMPER-SW2
WATER-PUMPU-CTRL
WATER-PUMP-SENSE-ADC
MAIN-BRUSH
MAIN-BRUSH-F-ADC
WHEEL-M-RIGHT-IN2
WHEEL-M-RIGHT-IN1
WHEEL-M-RIGHT-ENCODE-A
WHEEL-M-EN-V
WHEEL-M-LEFT-IN2
WHEEL-M-LEFT-IN1
WHEEL-M-LEFT-ENCODE-A
WHEEL-M-R-F-ADC
WHEEL-M-L-F-ADC
MAIN-FAN-V-CTRL
MAIN-FAN-S-CTRL
MAIN-FAN-S-SENSE
SIDE-BRUSH-V-R-CTRL
SIDE-BRUSH-V-L-CTRL
SIDE-BRUSH-R-F-ADC
SIDE-BRUSH-L-F-ADC
IMU-SPI-SCLK
IMU-SPI-MOSI
IMU-SPI-MISO
IMU-SPI-CS
IMU-INT#2
IMU-INT#1
IMU-FSYNC
LED-HOME
LED-PWR
HOME-BTN
PWR-BTN-SENSE
STM-PWR-CTRL
V-CHARGE-SENSE
VBAT-SENSE
CURENT-SENSE
CHARGE-STATUS
STM32-UART1-RX
STM32-UART1-TX
STM-RESET
RK-RESET
PMIC_PWRON
LiDAR-M-CTRL
LiDAR-TXD-HEADER

CONTROLLER-RK3562.SchDoc

RK3562

BUMPER-SW1
BUMPER-SW2
MIPI_CSI_RX0_D0P
MIPI_CSI_RX0_D0N
MIPI_CSI_RX0_D1P
MIPI_CSI_RX0_D1N
MIPI_CSI_RX0_D2P
MIPI_CSI_RX0_D2N
MIPI_CSI_RX0_D3P
MIPI_CSI_RX0_D3N
MIPI_CSI_RX0_CLK0P
MIPI_CSI_RX0_CLK0N
CAM_CLK0_OUT
I2C4_SCL_M0
I2C4_SDA_M0
CAM_RST0_L
CAM_PDN0_L
UART 9 _TX_M0
UART 9 _RX_M0
STM-RESET
RK-RESET
PMIC_PWRON
LiDAR-M-CTRL
USBC_P
USBC_N
LiDAR-TXD
LiDAR-RXD
SPK_P
SPK_N
AUDIO-HP-R
AUDIO-HP-L
AUDIO-SD
MIC1_IN
MIC2_IN
PMIC_EXT_EN
WIFI_PWREN_L
PMIC_32KOUT_WIFI
I2S2_SDI_M0
I2S2_SDO_M0
I2S2_LRCK_M0
I2S2_SCLK_M0
HOST_WAKE_BT_H
BT_REG_ON_H
UART1_RTSn_M0
UART1_CTSn_M0
UART1_TX_M0
UART1_RX_M0
WIFI_WAKE_HOST_H
WIFI_REG_ON_H
SDIO_CLK
SDIO_CMD
SDIO_D3
SDIO_D2
SDIO_D1
SDIO_D0
IMU-FSYNC-RK
IMU-INT2-RK
IMU-INT1-RK
IMU-SCLK
IMU-MOSI
IMU-MISO
IMU-CS

MIPI-CAMERA.SchDoc

CAMERA

MIPI_CSI_RX0_D0P
MIPI_CSI_RX0_D0N
MIPI_CSI_RX0_D1P
MIPI_CSI_RX0_D1N
MIPI_CSI_RX0_D2P
MIPI_CSI_RX0_D2N
MIPI_CSI_RX0_D3P
MIPI_CSI_RX0_D3N
MIPI_CSI_RX0_CLK0P
MIPI_CSI_RX0_CLK0N
CAM_CLK0_OUT
I2C4_SCL_M0
I2C4_SDA_M0
CAM_RST0_L
CAM_PDN0_L

LiDAR .SchDoc

LiDAR

LiDAR-RXD
LiDAR-MOTOR-CTRL

MAIN-FAN .SchDoc

MAIN FAN

MAIN-FAN-V-CTRL
MAIN-FAN-S-CTRL
MAIN-FAN-S-SENSE

SPEAKER-MIC.SchDoc

SPEAKER MIC

SPK_P
SPK_N
AUDIO-HP-R
AUDIO-HP-L
AUDIO-SD
MIC1_IN
MIC2_IN

SIDE-BRUSH-MOTORs .SchDoc

SIDE BURSH MOTORs

SIDE-BRUSH-V-R-CTRL
SIDE-BRUSH-V-L-CTRL
SIDE-BRUSH-R-F-ADC
SIDE-BRUSH-L-F-ADC

IMU-ICM-4267-P .SchDoc

IMU

IMU-SPI-SCLK
IMU-SPI-MOSI
IMU-SPI-MISO
IMU-SPI-CS
IMU-INT#2
IMU-INT#1
IMU-FSYNC

BMS-SYSTEM-POWER.SchDoc

POWER CONTROL

CHARGE-STATUS
CURENT-SENSE
VBAT-SENSE
V-CHARGE-SENSE
STM-PWR-CTRL
PWR-BTN-SENSE
POWER-BTN
USBC_P
USBC_N
PMIC_EXT_EN

BUTTON-LEDs .SchDoc

BUTTON-LEDs

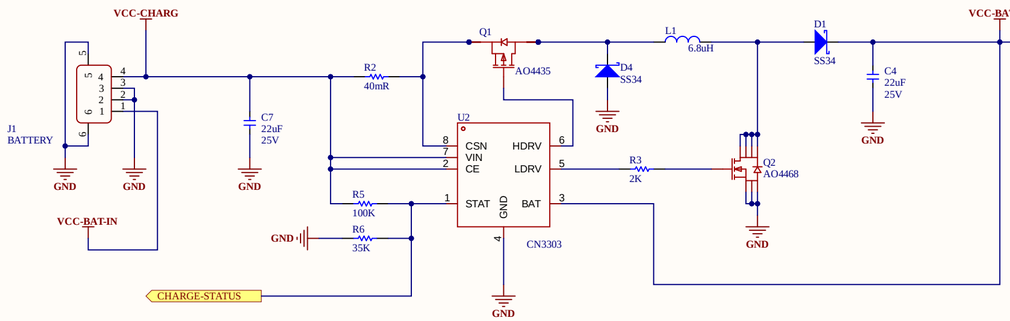
LED-HOME
LED-PWR
HOME-BTN
POWER-BTN

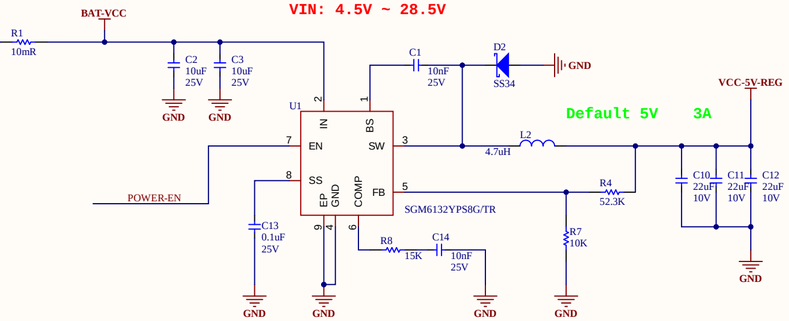
WF-BT-AP6256.SchDoc

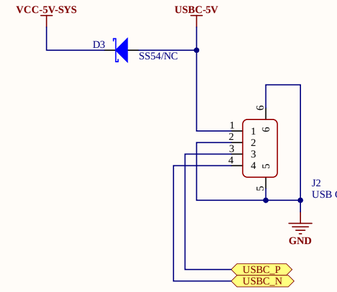
WF-BT-AP6256

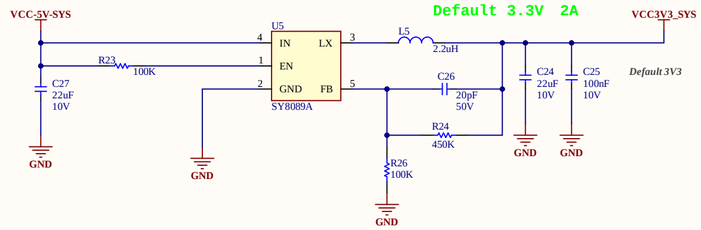
SDIO_D0
SDIO_D1
SDIO_D2
SDIO_D3
SDIO_CMD
SDIO_CLK
WIFI_REG_ON_H
WIFI_WAKE_HOST_H
UART1_RX_M0
UART1_TX_M0
UART1_RTSn_M0
UART1_CTSn_M0
BT_REG_ON_H
BT_WAKE_HOST_H
HOST_WAKE_BT_H
I2S2_SCLK_M0
I2S2_LRCK_M0
I2S2_SDO_M0
I2S2_SDI_M0
PMIC_32KOUT_WIFI
WIFI_PWREN_L

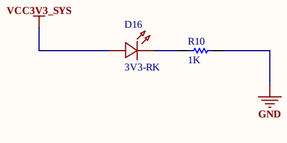
BATTERY CHARGE AND POWER MANAGMENT

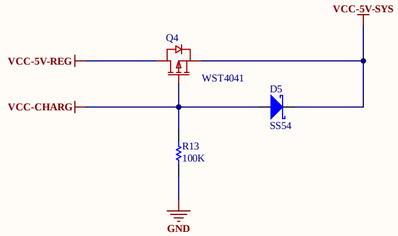


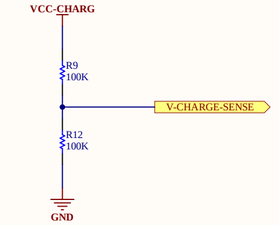


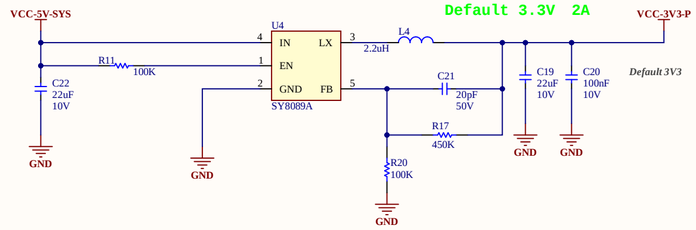


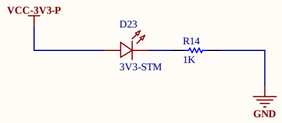


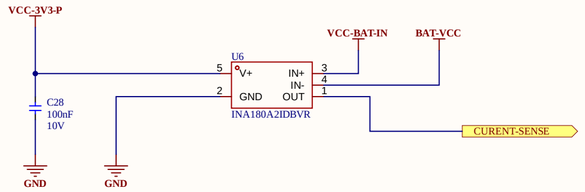


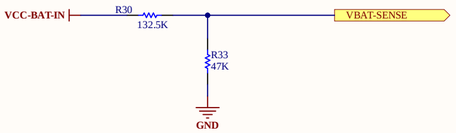


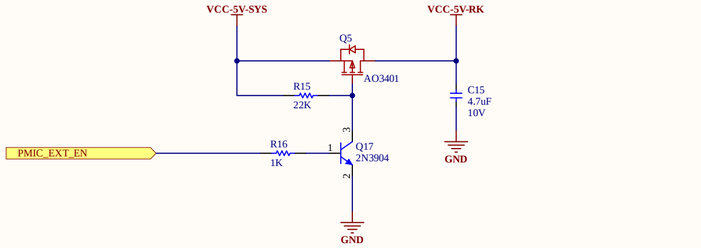


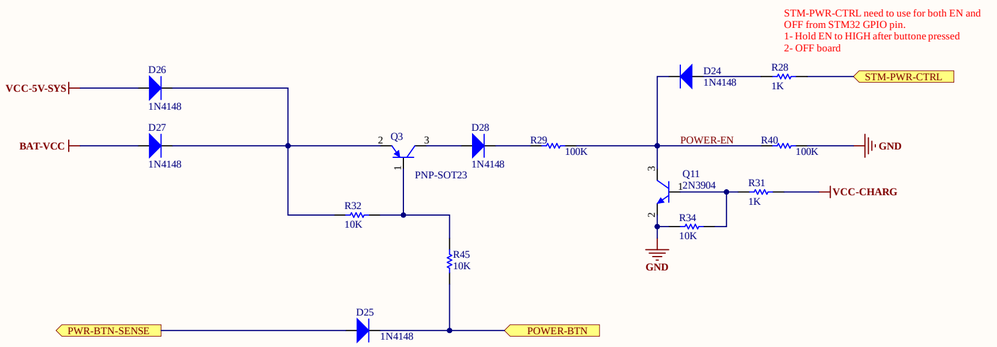












Controller-RK3562

PMIC_32KOUT_WIFI
PMIC_EXT_EN
PMIC_PWRON

RK-POWER-SUPPLY.SchDoc

RK-RESET
PMIC_EXT_EN
AUDIO-HP-R
AUDIO-HP-L
SPK_P
SPK_N
MIC1_IN
MIC2_IN

RESETn
PMIC_EXT_EN
HPR_OUT
HPL_OUT
HP_SNS
SPKP_OUT
SPKN_OUT
MIC1_IN
MIC2_IN

RK-PMIC-CODEC

I2C0_SCL_PMIC
I2C0_SDA_PMIC
PMIC_INT_L
PMIC_SLEEP_H
PMIC_PWRON
PMIC_32KOUT_WIFI
I2S0_MCLK_M0_PMIC
I2S0_SCLK_M0_PMIC
I2S0_LRCK_M0_PMIC
I2S0_SDI0_M0/PDM_SDI0_M0_PMIC
I2S0_SDO0_M0_PMIC
PDM_CLK0_M0_PMIC

RK-PERIPHERAL-POWER.SchDoc

PWM_NPU
CAM_PWREN_H

RK-PERIPH-POER

RK3562-VCCIO1.SchDoc

HP_DET_L_GPIO1_D0
I2S0_MCLK_M0_PMIC
I2S0_SCLK_M0_PMIC
I2S0_LRCK_M0_PMIC
I2S0_SDO0_M0_PMIC
PDM_CLK0_M0_PMIC
I2S0_SDI0_M0/PDM_SDI0_M0_PMIC

RK3562-VCCIO1

RK3562-VCCIO2-VCCIO3.SchDoc

RK3562-VCCIO2-VCCIO3

eMMC.SchDoc

eMMC_D0
eMMC_D1
eMMC_D2
eMMC_D3
eMMC_D4
eMMC_D5
eMMC_D6
eMMC_D7
eMMC_CMD
eMMC_CLK
eMMC_DATA_STROBE

eMMC

SD_TF-CARD.SchDoc

SDMMC0_D0
SDMMC0_D1
SDMMC0_D2
SDMMC0_D3
SDMMC0_CMD
SDMMC0_CLK
SDMMC0_DET_L

SD_TF-CARD

SD_TF-DET-L

RK3562-VCCIO4.SchDoc

SDIO_D0
SDIO_D1
SDIO_D2
SDIO_D3
SDIO_CMD
SDIO_CLK
UART1_TX_M0
UART1_RX_M0
UART1_RTSn_M0
UART1_CTSn_M0
I2S2_SCLK_M0
I2S2_LRCK_M0
I2S2_SDO_M0
I2S2_SDI_M0
SARADC0_BOOT
SARADC0_IN1_KEY/RECOVERY
SARADC0_IN4_HP_HOOK

RK3562-VCCIO4

RK3562-VCCIO5.SchDoc

MIPI_CSI_RX0_D0P
MIPI_CSI_RX0_D0N
MIPI_CSI_RX0_D1P
MIPI_CSI_RX0_D1N
MIPI_CSI_RX0_D2P
MIPI_CSI_RX0_D2N
MIPI_CSI_RX0_D3P
MIPI_CSI_RX0_D3N
MIPI_CSI_RX0_CLK0P
MIPI_CSI_RX0_CLK0N
CAM_CLK0_OUT
CAM_RST0_L
CAM_PDN0_L
I2C4_SCL_M0
I2C4_SDA_M0
CAM_PWREN0_H
FLASH_LED_EN
MIPI_CSI_RX1_D0P
MIPI_CSI_RX1_D0N
MIPI_CSI_RX1_D1P
MIPI_CSI_RX1_D1N
MIPI_CSI_RX1_CLK0P
MIPI_CSI_RX1_CLK0N
CAM_PDN1_L

RK3562-VCCIO5

RK3562-VCCIO6.SchDoc

MIPI_DSI_TX_D0P/LVDS_TX_D0P
MIPI_DSI_TX_D0N/LVDS_TX_D0N
MIPI_DSI_TX_D1P/LVDS_TX_D1P
MIPI_DSI_TX_D1N/LVDS_TX_D1N
MIPI_DSI_TX_D2P/LVDS_TX_D2P
MIPI_DSI_TX_D2N/LVDS_TX_D2N
MIPI_DSI_TX_D3P/LVDS_TX_D3P
MIPI_DSI_TX_D3N/LVDS_TX_D3N
MIPI_DSI_TX_CLKP/LVDS_TX_CLKP
MIPI_DSI_TX_CLKN/LVDS_TX_CLKN
UART4_TXD_M0
UART4_RXD_M0
SPK_STL
UART9_TXD_M0
UART9_RXD_M0
IMU-SCLK
IMU-MOSI
IMU-MISO
IMU-CS
IMU-FSYNC-RK
IMU-INT#1-RK
IMU-INT#2-RK
BUMPER-SW1
BUMPER-SW2
STM-RESET

RK3562-VCCIO6

LiDAR-TXD
LiDAR-RXD
AUDIO-SD
UART 9 _TX_M0
UART 9 _RX_M0
IMU-SCLK
IMU-MOSI
IMU-MISO
IMU-CS
IMU-FSYNC-RK
IMU-INT#1-RK
IMU-INT#2-RK
BUMPER-SW1
BUMPER-SW2
STM-RESET

Controller-RK3562-TOP[6C], RK3562-VCCIO6[4D]

RK3562-POWER-GND.SchDoc

RK3562-POWER-GND

RK3562-PMUIO-OSC-PLL.SchDoc

RESETn
SDMMC0_DET_L
PMIC_INT_L
I2C0_SCL_PMIC
I2C0_SDA_PMIC
PMIC_SLEEP_H
PWM_NPU
LCD_PWREN_H
PWM_GPU
LCD_BL_PWM
LCD_RST_L
TP_INT_L
TP_RST_L
I2C2_SCL_TP
I2C2_SDA_TP
USBCC_INT_L
WIFI_PWR
WIFI_REG_ON_H
WIFI_WAKE_HOST_H
BT_REG_ON_H
BT_WAKE_HOST_H
HOST_WAKE_BT_H
UART0_RX_M0_DEBUG
UART0_TX_M0_DEBUG

RK3562-PMUIO-OSC-PLL

RK-RESET
SD_TF-DET-L
PMIC_INT_L
I2C0_SCL_PMIC
I2C0_SDA_PMIC
PMIC_SLEEP_H
PWM_NPU
WIFI_PWREN_L
WIFI_REG_ON_H
WIFI_WAKE_HOST_H
BT_REG_ON_H
BT_WAKE_HOST_H
HOST_WAKE_BT_H

J3
RK3562-DEBUG
GND

RK3562-DDR-PHY.SchDoc

RK3562-DDR-PHY

LPDDR4_A0_A
LPDDR4_A1_A
LPDDR4_A2_A
LPDDR4_A3_A
LPDDR4_A4_A
LPDDR4_A5_A
LPDDR4_DQ0_A
LPDDR4_DQ1_A
LPDDR4_DQ2_A
LPDDR4_DQ3_A
LPDDR4_DQ4_A
LPDDR4_DQ5_A
LPDDR4_DQ6_A
LPDDR4_DQ7_A
LPDDR4_DQ8_A
LPDDR4_DQ9_A
LPDDR4_DQ10_A
LPDDR4_DQ11_A
LPDDR4_DQ12_A
LPDDR4_DQ13_A
LPDDR4_DQ14_A
LPDDR4_DQ15_A
LPDDR4_DQS0P_A
LPDDR4_DQS0N_A
LPDDR4_DQS1P_A
LPDDR4_DQS1N_A
LPDDR4_DM0_A
LPDDR4_DM1_A
LPDDR4_CLKP_A
LPDDR4_CLKN_A
LPDDR4_CKE0_A
LPDDR4_CKE1_A
LPDDR4_CS0n_A
LPDDR4_CS1n_A
LPDDR4_A0_B
LPDDR4_A1_B
LPDDR4_A2_B
LPDDR4_A3_B
LPDDR4_A4_B
LPDDR4_A5_B
LPDDR4_DQ0_B
LPDDR4_DQ1_B
LPDDR4_DQ2_B
LPDDR4_DQ3_B
LPDDR4_DQ4_B
LPDDR4_DQ5_B
LPDDR4_DQ6_B
LPDDR4_DQ7_B
LPDDR4_DQ8_B
LPDDR4_DQ9_B
LPDDR4_DQ10_B
LPDDR4_DQ11_B
LPDDR4_DQ12_B
LPDDR4_DQ13_B
LPDDR4_DQ14_B
LPDDR4_DQ15_B
LPDDR4_DQS0P_B
LPDDR4_DQS0N_B
LPDDR4_DQS1P_B
LPDDR4_DQS1N_B
LPDDR4_DM0_B
LPDDR4_DM1_B
LPDDR4_CLKP_B
LPDDR4_CLKN_B
LPDDR4_CKE0_B
LPDDR4_CKE1_B
LPDDR4_CS0n_B
LPDDR4_CS1n_B
LPDDR4_RESETn

RK3562-USB-PCIe-PHY.SchDoc

USB20_OTG_VBUSDET
USB20_OTG_DP
USB20_OTG_DM
USB20_OTG_ID

RK3562-USB-PCIe-PHY

USBC_P
USBC_N

DDR-LPDDR4.SchDoc

CHA

LPDDR4_A0_A
LPDDR4_A1_A
LPDDR4_A2_A
LPDDR4_A3_A
LPDDR4_A4_A
LPDDR4_A5_A
LPDDR4_DQ0_A
LPDDR4_DQ1_A
LPDDR4_DQ2_A
LPDDR4_DQ3_A
LPDDR4_DQ4_A
LPDDR4_DQ5_A
LPDDR4_DQ6_A
LPDDR4_DQ7_A
LPDDR4_DQ8_A
LPDDR4_DQ9_A
LPDDR4_DQ10_A
LPDDR4_DQ11_A
LPDDR4_DQ12_A
LPDDR4_DQ13_A
LPDDR4_DQ14_A
LPDDR4_DQ15_A
LPDDR4_DQS0P_A
LPDDR4_DQS0N_A
LPDDR4_DQS1P_A
LPDDR4_DQS1N_A
LPDDR4_DM0_A
LPDDR4_DM1_A
LPDDR4_CLKP_A
LPDDR4_CLKN_A
LPDDR4_CKE0_A
LPDDR4_CKE1_A
LPDDR4_CS0n_A
LPDDR4_CS1n_A

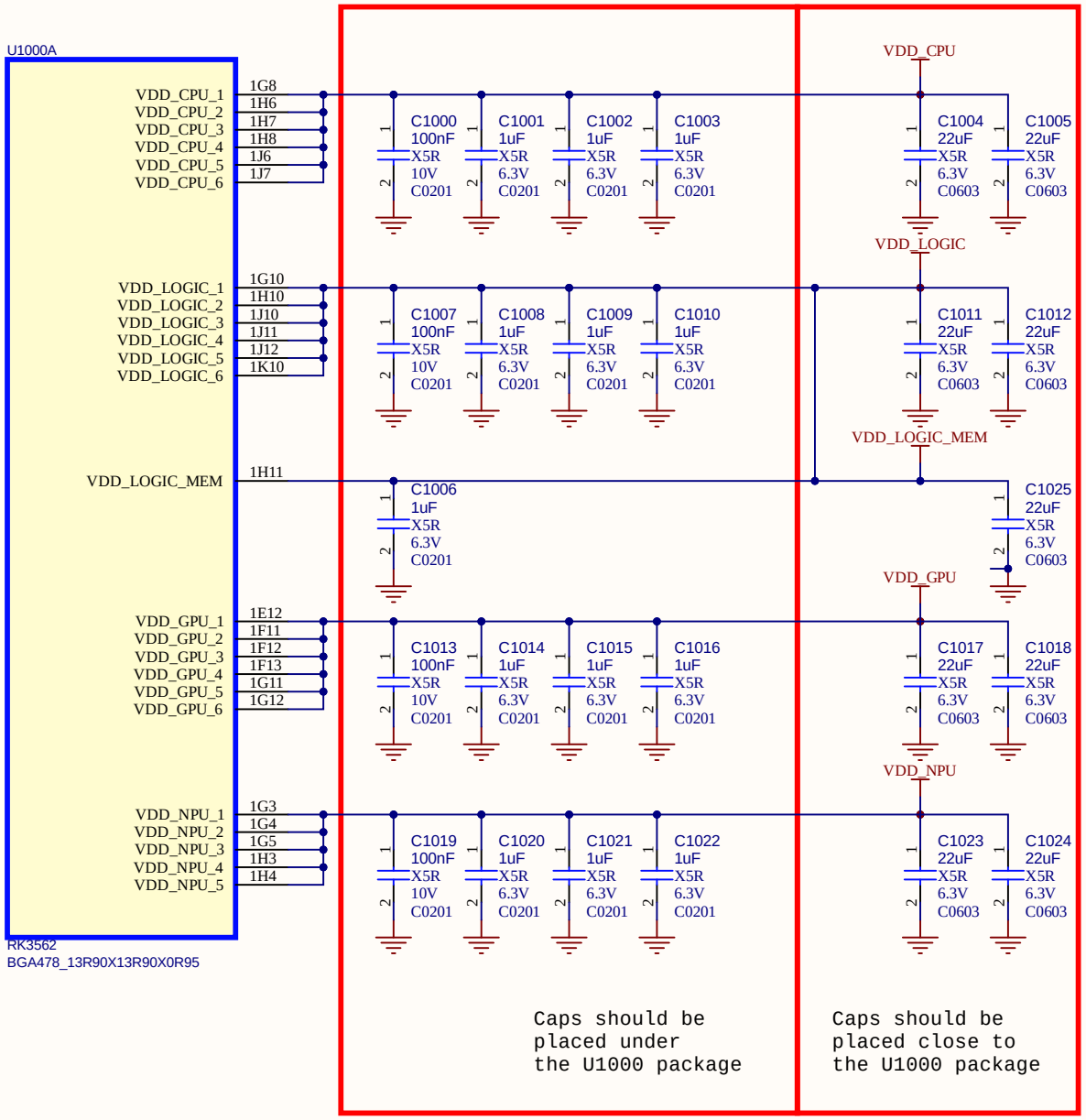
CHB

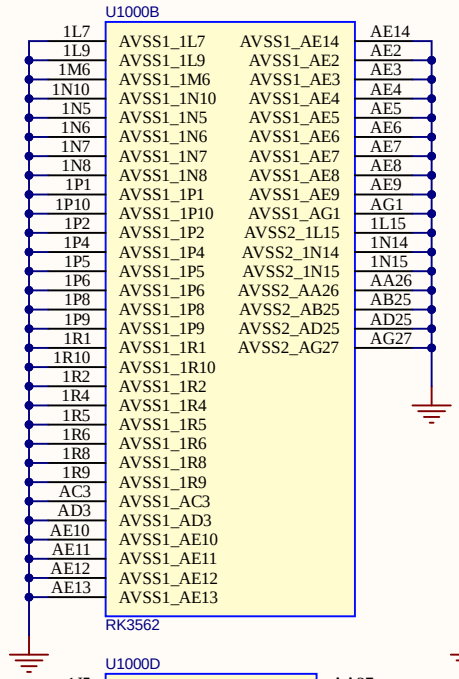
LPDDR4_A0_B
LPDDR4_A1_B
LPDDR4_A2_B
LPDDR4_A3_B
LPDDR4_A4_B
LPDDR4_A5_B
LPDDR4_DQ0_B
LPDDR4_DQ1_B
LPDDR4_DQ2_B
LPDDR4_DQ3_B
LPDDR4_DQ4_B
LPDDR4_DQ5_B
LPDDR4_DQ6_B
LPDDR4_DQ7_B
LPDDR4_DQ8_B
LPDDR4_DQ9_B
LPDDR4_DQ10_B
LPDDR4_DQ11_B
LPDDR4_DQ12_B
LPDDR4_DQ13_B
LPDDR4_DQ14_B
LPDDR4_DQ15_B
LPDDR4_DQS0P_B
LPDDR4_DQS0N_B
LPDDR4_DQS1P_B
LPDDR4_DQS1N_B
LPDDR4_DM0_B
LPDDR4_DM1_B
LPDDR4_CLKP_B
LPDDR4_CLKN_B
LPDDR4_CKE0_B
LPDDR4_CKE1_B
LPDDR4_CS0n_B
LPDDR4_CS1n_B
LPDDR4_RESETn

LPDDR4

RK3562_ABCD

(Power&GND)





RK3562_E(DDRPHY)

U1000E

DDR4	LPDDR4	DDR3	LPDDR3		
LPDDR4_DQ0_A	B6	DDR_DQ0_A	DDR4_DQL6_A	LPDDR4_DQ0_A	DDR3_DQ0	LPDDR3_DQ30
LPDDR4_DQ1_A	B8	DDR_DQ1_A	DDR4_DQL2_A	LPDDR4_DQ1_A	DDR3_DQ4	LPDDR3_DQ26
LPDDR4_DQ2_A	A8	DDR_DQ2_A	DDR4_DQL0_A	LPDDR4_DQ2_A	DDR3_DQ2	LPDDR3_DQ31
LPDDR4_DQ3_A	C8	DDR_DQ3_A	DDR4_DQL4_A	LPDDR4_DQ3_A	DDR3_DQ6	LPDDR3_DQ27
LPDDR4_DQ4_A	C5	DDR_DQ4_A	DDR4_DQL5_A	LPDDR4_DQ4_A	DDR3_DQ1	LPDDR3_DQ24
LPDDR4_DQ5_A	B5	DDR_DQ5_A	DDR4_DQL1_A	LPDDR4_DQ5_A	DDR3_DQ3	LPDDR3_DQ28
LPDDR4_DQ6_A	A6	DDR_DQ6_A	DDR4_DQL7_A	LPDDR4_DQ6_A	DDR3_DQ7	LPDDR3_DQ25
LPDDR4_DQ7_A	C6	DDR_DQ7_A	DDR4_DQL3_A	LPDDR4_DQ7_A	DDR3_DQ5	LPDDR3_DQ29
LPDDR4_DM0_A	B7	DDR_DM0_A	DDR4_DML_A	LPDDR4_DM0_A	DDR3_DM0	LPDDR3_DM3
LPDDR4_DQS0P_A	1A3	DDR_DQS0P_A	DDR4_DQSL_P_A	LPDDR4_DQS0P_A	DDR3_DQS0P	LPDDR3_DQS3P
LPDDR4_DQS0N_A	1B3	DDR_DQS0N_A	DDR4_DQSL_N_A	LPDDR4_DQS0N_A	DDR3_DQS0N	LPDDR3_DQS3N
LPDDR4_DQ8_A	C2	DDR_DQ8_A	DDR4_DQU1_A	LPDDR4_DQ8_A	DDR3_DQ14	LPDDR3_DQ9
LPDDR4_DQ9_A	C1	DDR_DQ9_A	DDR4_DQU5_A	LPDDR4_DQ9_A	DDR3_DQ10	LPDDR3_DQ13
LPDDR4_DQ10_A	B1	DDR_DQ10_A	DDR4_DQU3_A	LPDDR4_DQ10_A	DDR3_DQ12	LPDDR3_DQ15
LPDDR4_DQ11_A	B2	DDR_DQ11_A	DDR4_DQU7_A	LPDDR4_DQ11_A	DDR3_DQ8	LPDDR3_DQ14
LPDDR4_DQ12_A	A4	DDR_DQ12_A	DDR4_DQU0_A	LPDDR4_DQ12_A	DDR3_DQ13	LPDDR3_DQ12
LPDDR4_DQ13_A	B4	DDR_DQ13_A	DDR4_DQU4_A	LPDDR4_DQ13_A	DDR3_DQ15	LPDDR3_DQ8
LPDDR4_DQ14_A	A3	DDR_DQ14_A	DDR4_DQU2_A	LPDDR4_DQ14_A	DDR3_DQ11	LPDDR3_DQ11
LPDDR4_DQ15_A	B3	DDR_DQ15_A	DDR4_DQU6_A	LPDDR4_DQ15_A	DDR3_DQ9	LPDDR3_DQ10
LPDDR4_DM1_A	A2	DDR_DM1_A	DDR4_DMU_A	LPDDR4_DM1_A	DDR3_DM1	LPDDR3_DM1
LPDDR4_DQS1P_A	1A1	DDR_DQS1P_A	DDR4_DQSU_P_A	LPDDR4_DQS1P_A	DDR3_DQS1P	LPDDR3_DQS1P
LPDDR4_DQS1N_A	1B1	DDR_DQS1N_A	DDR4_DQSU_N_A	LPDDR4_DQS1N_A	DDR3_DQS1N	LPDDR3_DQS1N
LPDDR4_DQ0_B	B22	DDR_DQ0_B	DDR4_DQU7_B	LPDDR4_DQ0_B	DDR3_DQ16	LPDDR3_DQ17
LPDDR4_DQ1_B	B20	DDR_DQ1_B	DDR4_DQU3_B	LPDDR4_DQ1_B	DDR3_DQ20	LPDDR3_DQ21
LPDDR4_DQ2_B	A20	DDR_DQ2_B	DDR4_DQU1_B	LPDDR4_DQ2_B	DDR3_DQ22	LPDDR3_DQ16
LPDDR4_DQ3_B	C20	DDR_DQ3_B	DDR4_DQU5_B	LPDDR4_DQ3_B	DDR3_DQ18	LPDDR3_DQ20
LPDDR4_DQ4_B	C23	DDR_DQ4_B	DDR4_DQU0_B	LPDDR4_DQ4_B	DDR3_DQ21	LPDDR3_DQ23
LPDDR4_DQ5_B	B23	DDR_DQ5_B	DDR4_DQU4_B	LPDDR4_DQ5_B	DDR3_DQ23	LPDDR3_DQ19
LPDDR4_DQ6_B	C22	DDR_DQ6_B	DDR4_DQU2_B	LPDDR4_DQ6_B	DDR3_DQ19	LPDDR3_DQ18
LPDDR4_DQ7_B	A22	DDR_DQ7_B	DDR4_DQU6_B	LPDDR4_DQ7_B	DDR3_DQ17	LPDDR3_DQ22
LPDDR4_DM0_B	B21	DDR_DM0_B	DDR4_DMU_B	LPDDR4_DM0_B	DDR3_DM2	LPDDR3_DM2
LPDDR4_DQS0P_B	1A13	DDR_DQS0P_B	DDR4_DQSU_P_B	LPDDR4_DQS0P_B	DDR3_DQS2P	LPDDR3_DQS2P
LPDDR4_DQS0N_B	1B13	DDR_DQS0N_B	DDR4_DQSU_N_B	LPDDR4_DQS0N_B	DDR3_DQS2N	LPDDR3_DQS2N
LPDDR4_DQ8_B	C26	DDR_DQ8_B	DDR4_DQL0_B	LPDDR4_DQ8_B	DDR3_DQ26	LPDDR3_DQ0
LPDDR4_DQ9_B	C27	DDR_DQ9_B	DDR4_DQL4_B	LPDDR4_DQ9_B	DDR3_DQ30	LPDDR3_DQ1
LPDDR4_DQ10_B	B27	DDR_DQ10_B	DDR4_DQL2_B	LPDDR4_DQ10_B	DDR3_DQ28	LPDDR3_DQ2
LPDDR4_DQ11_B	B26	DDR_DQ11_B	DDR4_DQL6_B	LPDDR4_DQ11_B	DDR3_DQ24	LPDDR3_DQ6
LPDDR4_DQ12_B	A24	DDR_DQ12_B	DDR4_DQL5_B	LPDDR4_DQ12_B	DDR3_DQ25	LPDDR3_DQ3
LPDDR4_DQ13_B	B24	DDR_DQ13_B	DDR4_DQL7_B	LPDDR4_DQ13_B	DDR3_DQ31	LPDDR3_DQ5
LPDDR4_DQ14_B	A25	DDR_DQ14_B	DDR4_DQL1_B	LPDDR4_DQ14_B	DDR3_DQ27	LPDDR3_DQ4
LPDDR4_DQ15_B	B25	DDR_DQ15_B	DDR4_DQL3_B	LPDDR4_DQ15_B	DDR3_DQ29	LPDDR3_DQ7
LPDDR4_DM1_B	A26	DDR_DM1_B	DDR4_DML_B	LPDDR4_DM1_B	DDR3_DM3	LPDDR3_DM0
LPDDR4_DQS1P_B	1A15	DDR_DQS1P_B	DDR4_DQSL_P_B	LPDDR4_DQS1P_B	DDR3_DQS3P	LPDDR3_DQS0P
LPDDR4_DQS1N_B	1B15	DDR_DQS1N_B	DDR4_DQSL_N_B	LPDDR4_DQS1N_B	DDR3_DQS3N	LPDDR3_DQS0N

Note:
Except DDR3\DDR4, other DQ sequences
can not be swap

LPDDR3	DDR3	LPDDR4	DDR4		
--	DDR3_A9	LPDDR4_CLKN_B	DDR4_A0	AC0	A15 LPDDR4_CLKN_B
--	DDR3_A2	--	DDR4_A1	AC1	1B6
LPDDR3_A6	DDR3_A4	LPDDR4_A1_A	DDR4_A2	AC2	A10 LPDDR4_A1_A
LPDDR3_ODT0	DDR3_ODT1	LPDDR4_CS0N_A	DDR4_A3	AC3	B11 LPDDR4_CS0n_A
--	DDR3_BA1	LPDDR4_A3_B	DDR4_A4	AC4	1A10 LPDDR4_A3_B
--	DDR3_A11	LPDDR4_A5_B	DDR4_A5	AC5	1A8 LPDDR4_A5_B
LPDDR3_A1	DDR3_A14	LPDDR4_A1_B	DDR4_A6	AC6	A18 LPDDR4_A1_B
LPDDR3_A0	DDR3_A5	LPDDR4_A0_B	DDR4_A7	AC7	B18 LPDDR4_A0_B
LPDDR3_A9	DDR3_A7	LPDDR4_ODT0_CA_A	DDR4_A8	AC8	B9
--	DDR3_A13	LPDDR4_CLKP_B	DDR4_A9	AC9	B15 LPDDR4_CLKP_B
LPDDR3_CS0N	DDR3_A10	LPDDR4_CKE0_B	DDR4_A10	AC10	C16 LPDDR4_CKE0_B
LPDDR3_A8	DDR3_A6	LPDDR4_A0_A	DDR4_A11	AC11	B10 LPDDR4_A0_A
--	DDR3_BA0	LPDDR4_A3_A	DDR4_A12	AC12	1A6 LPDDR4_A3_A
LPDDR3_ODT1	DDR3_A8	LPDDR4_ODT0_CA_B	DDR4_A13	AC13	B19
--	DDR3_A15	LPDDR4_A4_A	DDR4_A14_WEN	AC14	1B8 LPDDR4_A4_A
LPDDR3_A7	DDR3_A0	LPDDR4_A2_A	DDR4_A15_CASN	AC15	C10 LPDDR4_A2_A
--	DDR3_RASN	LPDDR4_A5_A	DDR4_A16_RASN	AC16	1A7 LPDDR4_A5_A
LPDDR3_CS1N	DDR3_CASN	LPDDR4_CKE1_B	DDR4_ACTN	AC17	B16 LPDDR4_CKE1_B
LPDDR3_A3	DDR3_A1	LPDDR4_A2_B	DDR4_BA0	AC18	C18 LPDDR4_A2_B
--	DDR3_A12	LPDDR4_A4_B	DDR4_BA1	AC19	1A9 LPDDR4_A4_B
--	DDR3_WEN	LPDDR4_ODT1_CA_B	DDR4_BG0	AC20	1B10
--	DDR3_BA2	LPDDR4_ODT1_CA_A	DDR4_BG1	AC21	1B7
LPDDR3_CKE	DDR3_CKE	LPDDR4_CKE0_A	DDR4_CKE0	AC22	C12 LPDDR4_CKE0_A
LPDDR3_CLKP	DDR3_CLKP	LPDDR4_CLKN_A	DDR4_CLK0P	AC23	A13 LPDDR4_CLKN_A
LPDDR3_CLKN	DDR3_CLKN	LPDDR4_CLKP_A	DDR4_CLK0N	AC24	B13 LPDDR4_CLKP_A
--	DDR3_A3	LPDDR4_CKE1_A	DDR4_CS0N	AC25	B12
LPDDR3_A5	DDR3_CS1N	LPDDR4_CS1N_A	DDR4_CS1N	AC26	C11 LPDDR4_CKE1_A
LPDDR3_A4	DDR3_ODT0	LPDDR4_CS1N_B	DDR4_ODT0	AC27	C17 LPDDR4_CS1n_A
LPDDR3_A2	DDR3_CS0N	LPDDR4_CS0N_B	DDR4_ODT1	AC28	B17 LPDDR4_CS1n_B
--	DDR3_RESETN	LPDDR4_RESETN	DDR4_RESETN	AC29	1B11 LPDDR4_CS0n_B

Note: Sequences can not be swap. Note the CLK polarity.

	VDDQ	VDDQL
DDR3L	1.35V	1.35V
DDR3	1.50V	1.50V
DDR4	1.20V	1.20V
LPDDR3	1.20V	1.20V
LPDDR4	1.10V	1.10V
LPDDR4X	1.10V	**0.60V**

RK3562

DDR_RZQ — 1D4 — DDR_RZQ R1101 1 — 2 120R 1% R0201 — VCC_DDR

USE LP4/LP4X, R1101 PULL UP TO VCC_DDR

DDR_VREFOUT — 1C9

VCC_DDR

DDR_VDDQ_1	1E3	C1108 100nF X5R 10V C0201	C1100 100nF X5R 10V C0201	C1101 100nF X5R 10V C0201	C1102 4.7uF X5R 6.3V C0402	C1103 10uF X5R 6.3V C0603
DDR_VDDQ_2	1E5					
DDR_VDDQ_3	1D6					
DDR_VDDQ_4	1E7					
DDR_VDDQ_5	1E9					
DDR_VDDQ_6	1D11					

VCC0V6_DDR

DDR_VDDQL_1	1E4	C1109 100nF X5R 10V C0201	C1104 100nF X5R 10V C0201	C1105 100nF X5R 10V C0201	C1106 4.7uF X5R 6.3V C0402	C1107 4.7uF X5R 6.3V C0402
DDR_VDDQL_2	1E6					
DDR_VDDQL_3	1E8					
DDR_VDDQL_4	1E10					

Caps should be
placed under
the U1000 package

RK3562_F(PMUIO/OSC/PLL)

Adjust the load capacitor
according to the crystal spec.

U1000F

OSC

R1200 1 22R 2 R0201 5% XOUT24M AF25 → OSC_SOC_XOUT

Y1200 24MHz (X1, GND, GND, X2) CRY4_3R20X2R50X0R80

R1201 1M 5% R0201

C1201 18pF C0G 25V C0201

C1202 18pF C0G 25V C0201

XIN24M AG25 → OSC_SOC_XIN

1N13 → TVSS

PMU PLL

VDDA0V9_PMU, PMUPLL_AVDD_0V9

R1236 1 0R 2 R0402 5%

C1205 100nF X5R 10V C0201

1M13 → PMUPLL_AVDD_0V9

VCCA1V8_PMU, PMUPLL_AVDD_1V8

R1237 1 0R 2 R0402 5%

C1207 100nF X5R 10V C0201

1L13 → PMUPLL_AVDD_1V8

1M12 → PMUPLL_AVSS

VDDA_0V9, SYSPLL_AVDD_0V9

R1238 1 0R 2 R0402 5%

C1210 100nF X5R 10V C0201

1M10 → SYSPLL_AVDD_0V9

SYS PLL

VCC_1V8, SYSPLL_AVDD_1V8

R1242 1 0R 2 R0402 5%

C1212 100nF X5R 10V C0201

1G1 → SYSPLL_AVDD_1V8

Caps should be
placed under
the U1000 package

RK3562

PMUIO0 Domain

Operating Voltage=3.3V Only

nPOR 1P12

PMUIO0_3V3 → R0402 R1203 10K

Caps should be
placed under
the U1000 package

C1200 100nF X5R 10V C0201

RESETn

--	TSADC_SHUT_ORG	TSADC_SHUT_M0	GPIO0_A1_z	1R13 TSADC_SHUT_M0 R1202 1 0R 2 R0201 5%
--	--	REF_CLK_OUT	GPIO0_A0_d	AE24 → WIFI_PWR
--	TSADC_SHUT_M1	PWR_CTRL0	GPIO0_A2_d	AF23 → PMIC_SLEEP_H
--	--	PWR_CTRL1	GPIO0_A3_u	AF22 → PMIC_INT_L
--	I2C4_SDA_M1	SDMMC0_DETN	GPIO0_A4_u	1R15 → SDMMC0_DET_L
PMU_DEBUG	I2C4_SCL_M1	SDMMC0_PWREN	GPIO0_A5_d	AE22
--	--	PCIE20_CLKREQN_M0	GPIO0_A6_d	AG24
--	--	PWM3_M0	GPIO0_A7_d	AG21 → USBCC_INT_L
PCIE20_BUTTONRSTN	CLK0_32K_OUT	CLK_32K_IN	GPIO0_B0_d	AF21 → LCD_PWREN_H
--	--	I2C0_SCL	GPIO0_B1_d	AG23 → I2C0_SCL_PMIC
--	--	I2C0_SDA	GPIO0_B2_d	AE23 → I2C0_SDA_PMIC

PMUIO0 1L12

PMUIO0_3V3, VCC3V3_PMU

R1235 1 0R 2 R0402 5%

C1203 100nF X5R 10V C0201

Caps should be
placed under
the U1000 package

PMUIO1 Domain

Operating Voltage=1.8V/3.3V

--	--	--	I2C1_SCL_M0	GPIO0_B3_d
--	--	--	I2C1_SDA_M0	GPIO0_B4_d
--	--	PCIE20_PERSTN_M0	I2C2_SCL_M0	GPIO0_B5_d
--	--	PCIE20_WAKEN_M0	I2C2_SDA_M0	GPIO0_B6_d
SPDIF_TX_M1	CPU_AVS	PWM4_M0	SPI0_CSN1_M0	GPIO0_B7_d
CAN1_TX_M1	GPU_AVS	PWM7_M0	UART2_TX_M0	GPIO0_C0_d
CAN1_RX_M1	NPU_AVS	PWM6_M0	UART2_RX_M0	GPIO0_C1_d
--	SPI0_CSN0_M0	PWM5_M0	UART2_CTSN_M0	GPIO0_C2_d
--	SPI0_CLK_M0	PWM0_M0	UART2_RTSN_M0	GPIO0_C3_d
--	SPI0_MOSI_M0	PWM1_M0	UART6_CTSN_M0	GPIO0_C4_d
--	SPI0_MISO_M0	PWM2_M0	UART6_RTSN_M0	GPIO0_C5_d
--	--	CAN0_TX_M2	UART6_TX_M0	GPIO0_C6_d
--	--	CAN0_RX_M2	UART6_RX_M0	GPIO0_C7_d

AE17 → WIFI_REG_ON_H
AF18 → WIFI_WAKE_HOST_H
AF16 → I2C2_SCL_TP
AE15 → I2C2_SDA_TP
AF15 → TP_RST_L
AF17 → PWM_GPU
1R11 → PWM_NPU
AG17 → LCD_BL_PWM
AE16 → TP_INT_L
AG19 → LCD_RST_L
AE18 → BT_REG_ON_H
AF19 → BT_WAKE_HOST_H
AE19 → HOST_WAKE_BT_H

--	--	JTAG_CPU_MCU_TMS_M0	UART0_RX_M0	GPIO0_D0_u
--	--	JTAG_CPU_MCU_TCK_M0	UART0_TX_M0	GPIO0_D1_u

AF20 → UART0_RX_M0_DEBUG
AE20 → UART0_TX_M0_DEBUG

PMUIO1 1L11

PMUIO1, VCCA1V8_PMU, VCC3V3_PMU

R1239 1 0R 2 R0402 5%

C1213 100nF X5R 10V C0201

R41 1 0R/NC 2 R0402 5%

PMUIO0/1/OSC Domain Logic Power

Operating Voltage=0.9V

PMU_VDD_LOGIC_0V9 1M11

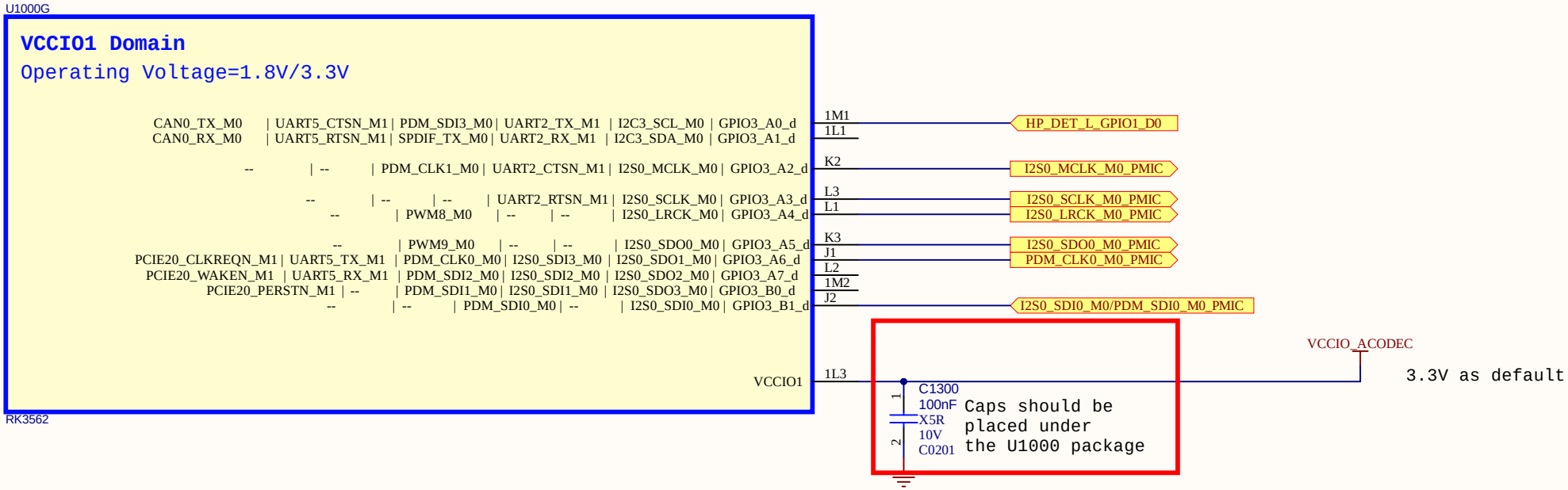
VDD_PMU_LOGIC_0V9, VDDA0V9_PMU

R1243 1 0R 2 R0402 5%

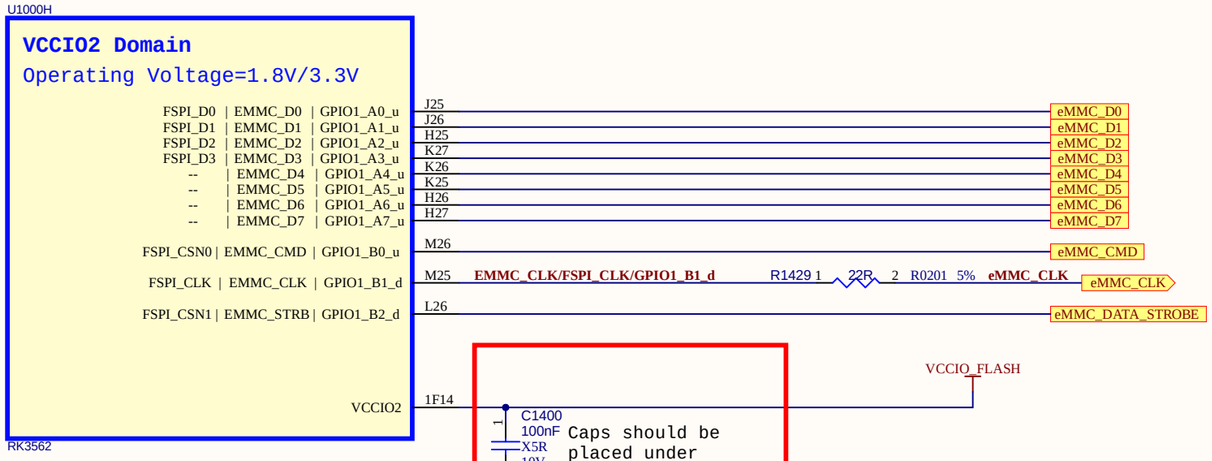
C1214 100nF X5R 10V C0201

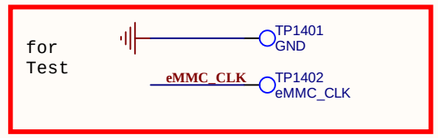
Caps should be
placed under
the U1000 package

RK3562_G(VCCIO1 Domain)

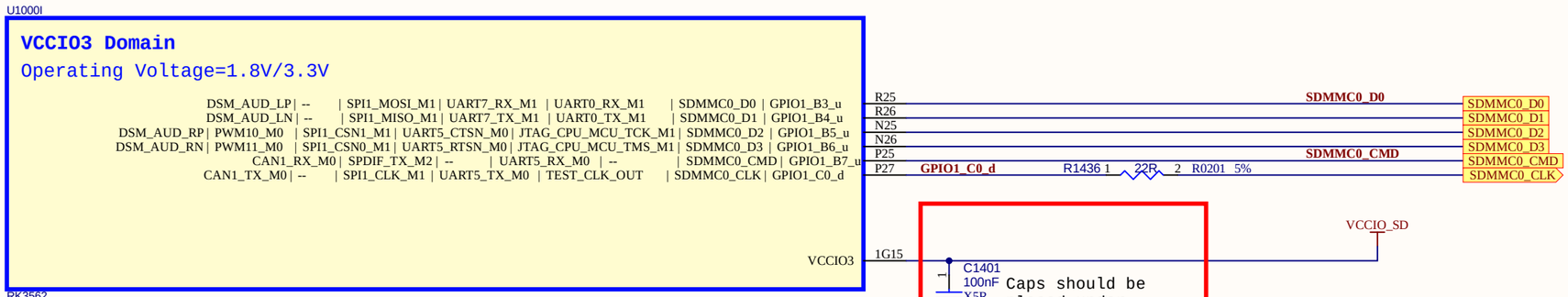


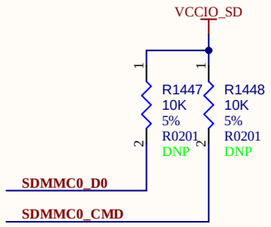
RK3562_H(VCCIO2 Domain)



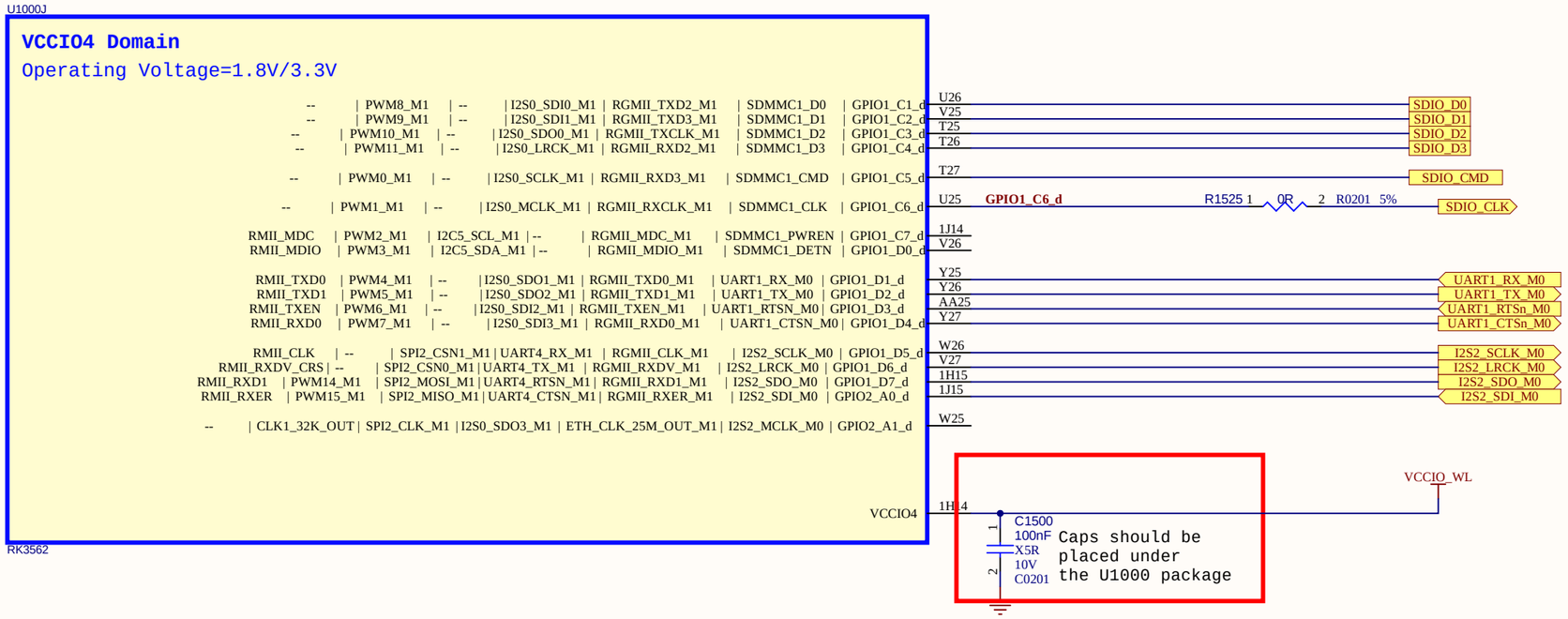


RK3562_I(VCCIO3 Domain)





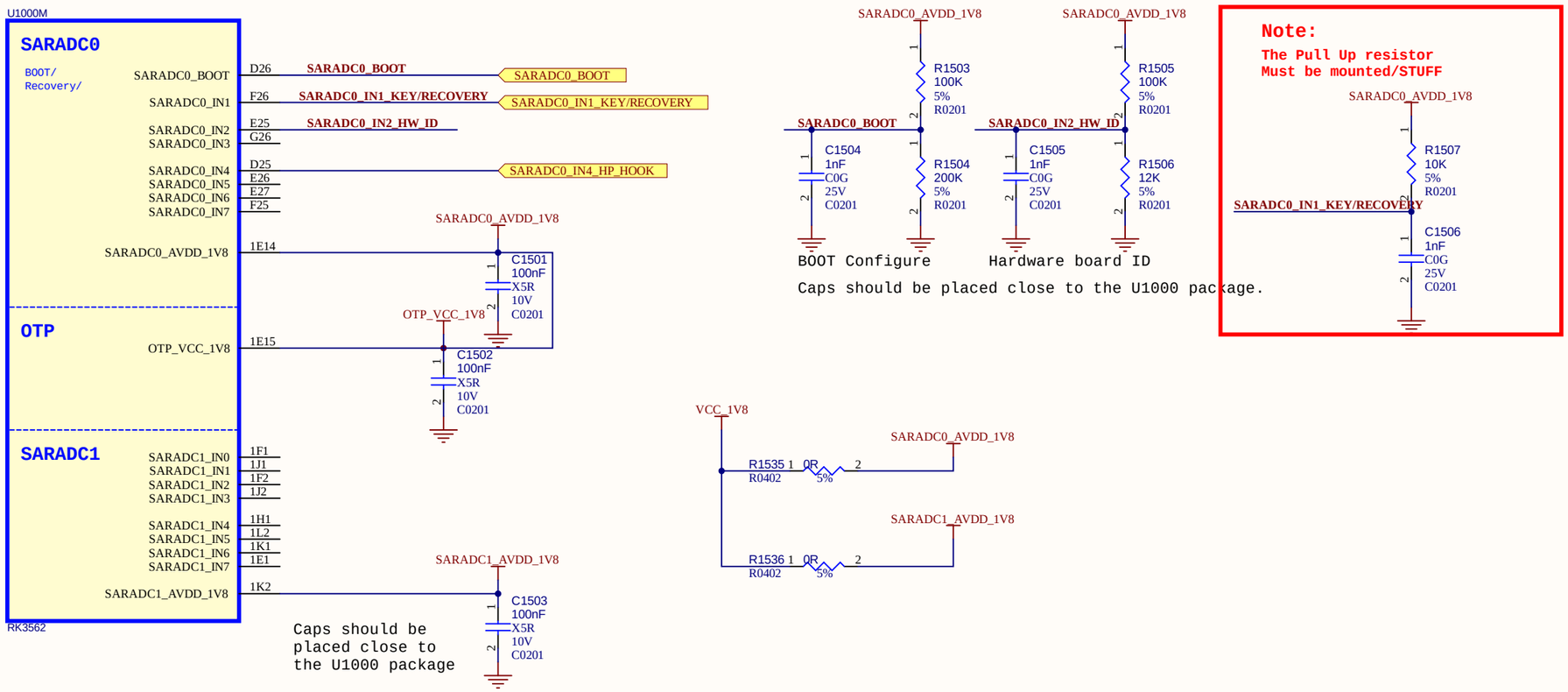
RK3562_J(VCCIO4 Domain)



Config Table for SARADC0_BOOT

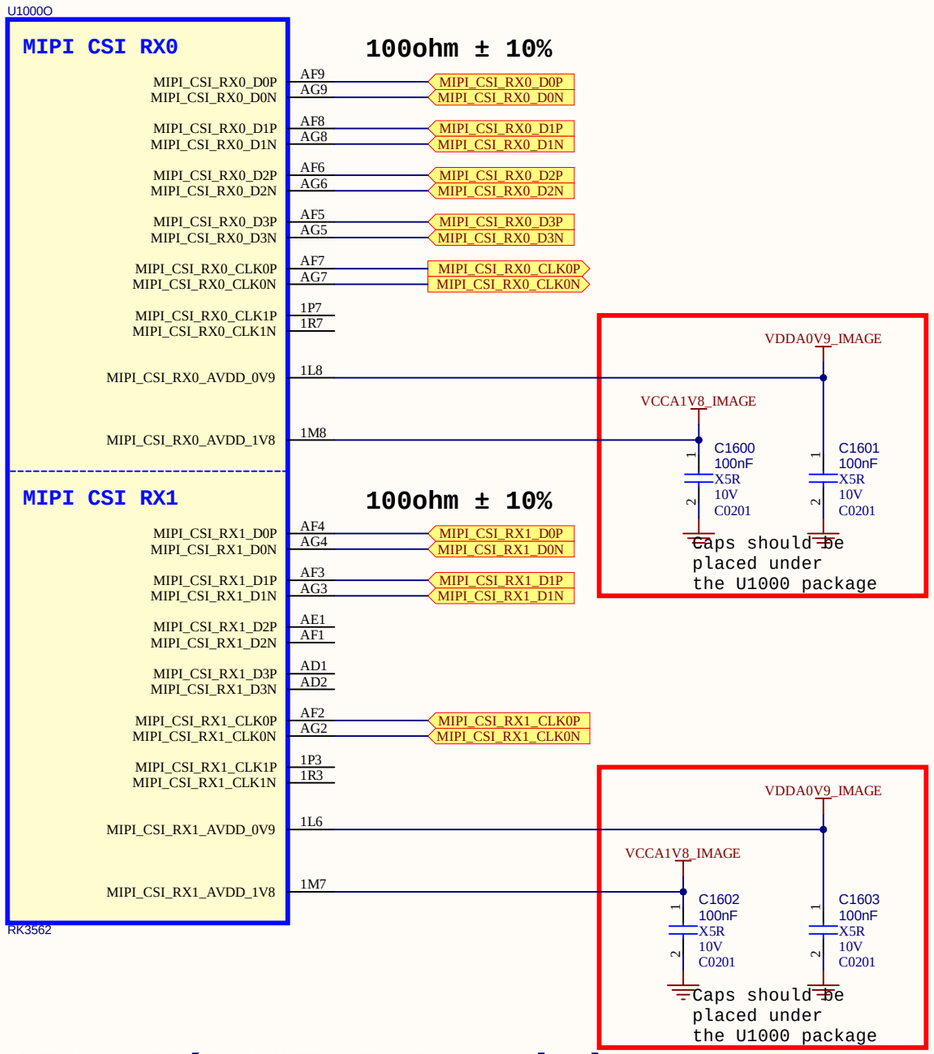
Item	*Rup*	*Rdown*	*ADC Value*	*Boot Mode*
Config1	NC	100K	0	USB (Maskrom mode)
Config2	100K	12K	114	
Config3	100K	27K	228	FSPI--USB
Config4	100K	51K	342	
Config5	100K	82K	456	
Config6	100K	120K	570	EMMC--USB
Config7	100K	200K	683	EMMC--SD Card--USB
Config8	100K	330K	796	SD Card--USB
Config9	100K	820K	910	
Config10	100K	NC	1023	FSPI--EMMC--SD Card--USB

RK3562_M(SARADC/OTP)

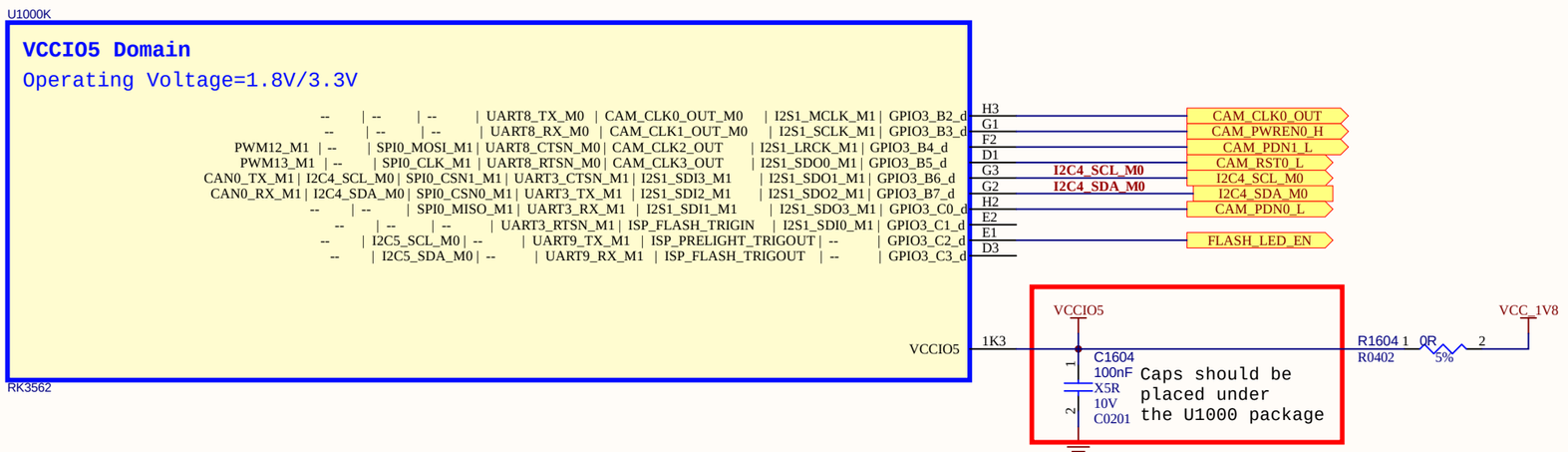


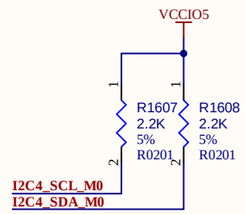
RK3562_O(MIPI_CSI_RX0 & RX1)

Usage of each MIPI CSI RXn & CLKn		
Option1	Sensor1 x4Lane	MIPI_CSI_RXn_D0-3 MIPI_CSI_RXn_CLK0
Option2	Sensor1 x2Lane + Sensor2 x2Lane	MIPI_CSI_RXn_D0-1 MIPI_CSI_RXn_CLK0 MIPI_CSI_RXn_D2-3 MIPI_CSI_RXn_CLK1



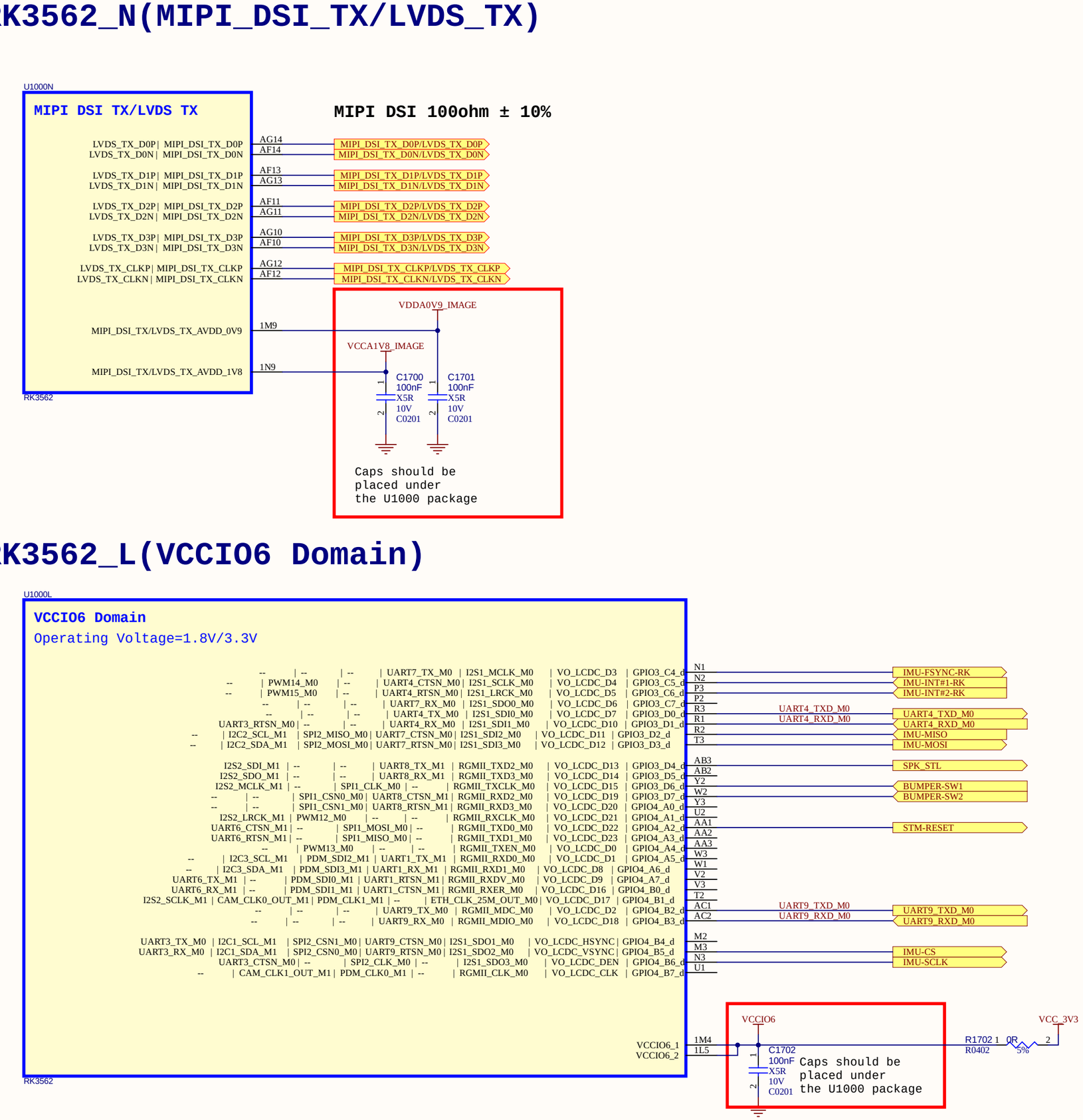
RK3562_K(VCCIO5 Domain)



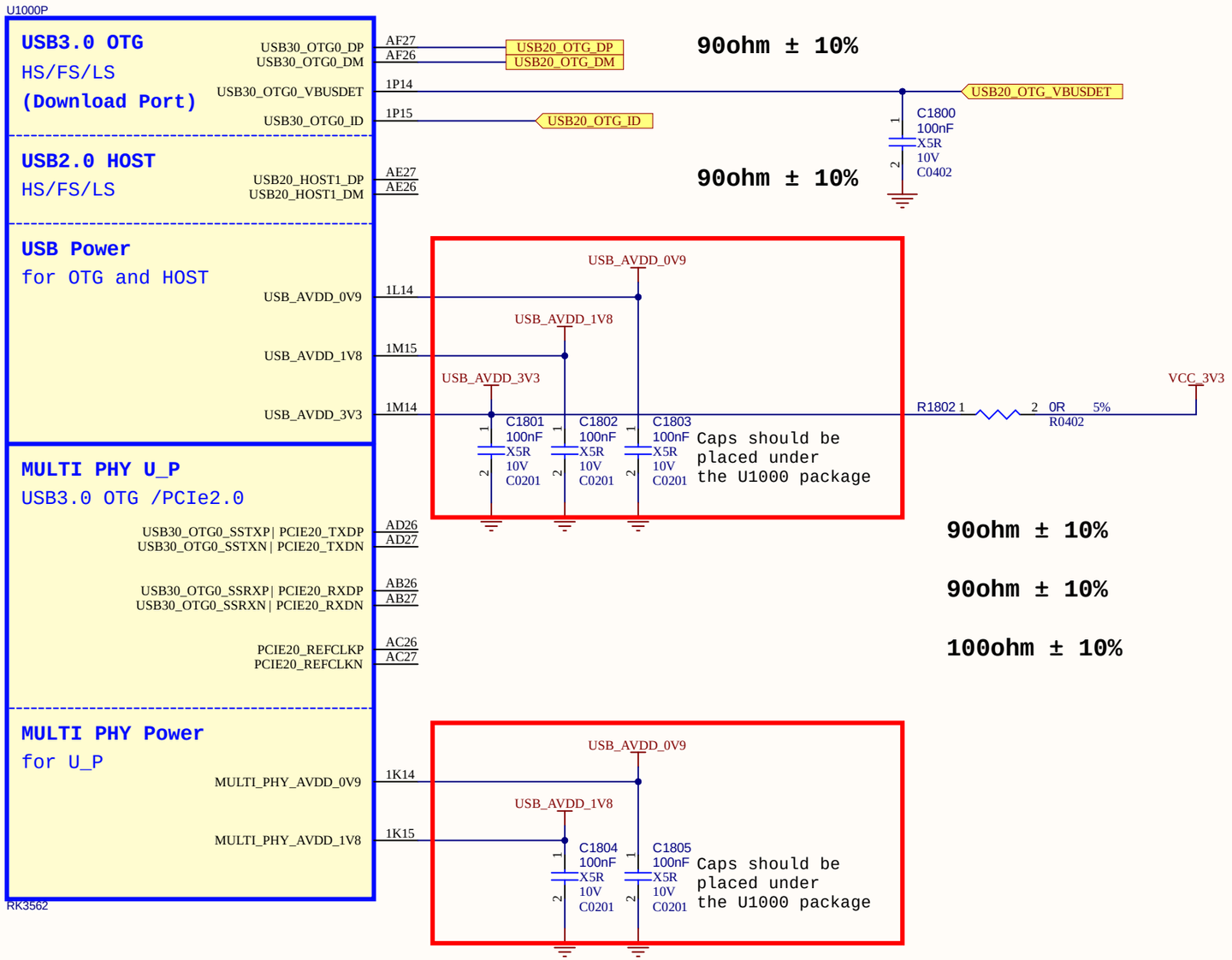


RK3562_N(MIPI_DSI_TX/LVDS_TX)

RK3562_L(VCCIO6 Domain)



RK3562_P(USB/PCIe)



LPDDR4

U3000A

CH A / CH B

LPDDR4_DQ0_A	B2	DQ0_a	DQ0_b	AA2	LPDDR4_DQ0_B
LPDDR4_DQ1_A	C2	DQ1_a	DQ1_b	Y2	LPDDR4_DQ1_B
LPDDR4_DQ2_A	E2	DQ2_a	DQ2_b	V2	LPDDR4_DQ2_B
LPDDR4_DQ3_A	F2	DQ3_a	DQ3_b	U2	LPDDR4_DQ3_B
LPDDR4_DQ4_A	F4	DQ4_a	DQ4_b	U4	LPDDR4_DQ4_B
LPDDR4_DQ5_A	E4	DQ5_a	DQ5_b	V4	LPDDR4_DQ5_B
LPDDR4_DQ6_A	C4	DQ6_a	DQ6_b	Y4	LPDDR4_DQ6_B
LPDDR4_DQ7_A	B4	DQ7_a	DQ7_b	AA4	LPDDR4_DQ7_B
LPDDR4_DQS0P_A	D3	DQS0_t_a	DQS0_t_b	W3	LPDDR4_DQS0P_B
LPDDR4_DQS0N_A	E3	DQS0_c_a	DQS0_c_b	V3	LPDDR4_DQS0N_B
LPDDR4_DM0_A	C3	DMI0_a	DMI0_b	Y3	LPDDR4_DM0_B
LPDDR4_DQ8_A	B11	DQ8_a	DQ8_b	AA11	LPDDR4_DQ8_B
LPDDR4_DQ9_A	C11	DQ9_a	DQ9_b	Y11	LPDDR4_DQ9_B
LPDDR4_DQ10_A	E11	DQ10_a	DQ10_b	V11	LPDDR4_DQ10_B
LPDDR4_DQ11_A	F11	DQ11_a	DQ11_b	U11	LPDDR4_DQ11_B
LPDDR4_DQ12_A	F9	DQ12_a	DQ12_b	U9	LPDDR4_DQ12_B
LPDDR4_DQ13_A	E9	DQ13_a	DQ13_b	V9	LPDDR4_DQ13_B
LPDDR4_DQ14_A	C9	DQ14_a	DQ14_b	Y9	LPDDR4_DQ14_B
LPDDR4_DQ15_A	B9	DQ15_a	DQ15_b	AA9	LPDDR4_DQ15_B
LPDDR4_DQS1P_A	D10	DQS1_t_a	DQS1_t_b	W10	LPDDR4_DQS1P_B
LPDDR4_DQS1N_A	E10	DQS1_c_a	DQS1_c_b	V10	LPDDR4_DQS1N_B
LPDDR4_DM1_A	C10	DMI1_a	DMI1_b	Y10	LPDDR4_DM1_B
LPDDR4_A0_A	H2	CA0_a	CA0_b	R2	LPDDR4_A0_B
LPDDR4_A1_A	J2	CA1_a	CA1_b	P2	LPDDR4_A1_B
LPDDR4_A2_A	H9	CA2_a	CA2_b	R9	LPDDR4_A2_B
LPDDR4_A3_A	H10	CA3_a	CA3_b	R10	LPDDR4_A3_B
LPDDR4_A4_A	H11	CA4_a	CA4_b	R11	LPDDR4_A4_B
LPDDR4_A5_A	J11	CA5_a	CA5_B	P11	LPDDR4_A5_B
LPDDR4_CLKP_A	J8	CK_t_a	CK_t_b	P8	LPDDR4_CLKP_B
LPDDR4_CLKN_A	J9	CK_c_a	CK_c_b	P9	LPDDR4_CLKN_B
LPDDR4_CKE0_A	J4	CKE0_a	CKE0_b	P4	LPDDR4_CKE0_B
LPDDR4_CKE1_A	J5	CKE1_a	CKE1_b	P5	LPDDR4_CKE1_B
	K8	CKE2_a_NC	CKE2_b_NC	N8	
LPDDR4_CS0n_A	H4	CS0_a	CS0_b	R4	LPDDR4_CS0n_B
LPDDR4_CS1n_A	H3	CS1_a	CS1_b	R3	LPDDR4_CS1n_B
	K5	CS2_a_NC	CS2_b_NC	N5	
VCC_DDR	G2	ODT_CA_a	ODT_CA_b	T2	VCC_DDR
	A5	ZQ0			
	A8	ZQ1			
	G11	ZQ2_NC	RESET_n	T11	LPDDR4_RESETn

LPDDR4_200P
BGA200_15R00X10R00X0R90

VCC0V6_DDR
R3001 240R R0201 1%
R3002 240R R0201 1%

C3001 1nF C0402 X5R 50V DNP

U3000B

VCC1V8_DDR	F1	VDD1_1	
	F12	VDD1_2	
	G4	VDD1_3	
	G9	VDD1_4	
	T4	VDD1_5	
	T9	VDD1_6	
	U1	VDD1_7	
	U12	VDD1_8	
VCC_DDR	A4	VDD2_1	
	A9	VDD2_2	
	F5	VDD2_3	
	F8	VDD2_4	
	H1	VDD2_5	
	H5	VDD2_6	
	H8	VDD2_7	
	H12	VDD2_8	
	K1	VDD2_9	
	K3	VDD2_10	
	K10	VDD2_11	
	K12	VDD2_12	
	N1	VDD2_13	
	N3	VDD2_14	
	N10	VDD2_15	
	N12	VDD2_16	
	R1	VDD2_17	
	R5	VDD2_18	
	R8	VDD2_19	
	R12	VDD2_20	
	U5	VDD2_21	
	U8	VDD2_22	
	AB4	VDD2_23	
	AB9	VDD2_24	
VCC0V6_DDR	B3	VDDQ_1	
	B5	VDDQ_2	
	B8	VDDQ_3	
	B10	VDDQ_4	
	D1	VDDQ_5	
	D5	VDDQ_6	
	D8	VDDQ_7	
	D12	VDDQ_8	
	F3	VDDQ_9	
	F10	VDDQ_10	
	U3	VDDQ_11	
	U10	VDDQ_12	
	W1	VDDQ_13	
	W5	VDDQ_14	
	W8	VDDQ_15	
	W12	VDDQ_16	
	AA3	VDDQ_17	
	AA5	VDDQ_18	
	AA8	VDDQ_19	
	AA10	VDDQ_20	
		VSS_1	A3
		VSS_2	A10
		VSS_3	C1
		VSS_4	C5
		VSS_5	C8
		VSS_6	C12
		VSS_7	D2
		VSS_8	D4
		VSS_9	D9
		VSS_10	D11
		VSS_11	E1
		VSS_12	E5
		VSS_13	E8
		VSS_14	E12
		VSS_15	G1
		VSS_16	G3
		VSS_17	G5
		VSS_18	G8
		VSS_19	G10
		VSS_20	G12
		VSS_21	J1
		VSS_22	J3
		VSS_23	J10
		VSS_24	J12
		VSS_25	K2
		VSS_26	K4
		VSS_27	K9
		VSS_28	K11
		VSS_29	N2
		VSS_30	N4
		VSS_31	N9
		VSS_32	N11
		VSS_33	P1
		VSS_34	P3
		VSS_35	P10
		VSS_36	P12
		VSS_37	T1
		VSS_38	T3
		VSS_39	T5
		VSS_40	T8
		VSS_41	T10
		VSS_42	T12
		VSS_43	V1
		VSS_44	V5
		VSS_45	V8
		VSS_46	V12
		VSS_47	W2
		VSS_48	W4
		VSS_49	W9
		VSS_50	W11
		VSS_51	Y1
		VSS_52	Y5
		VSS_53	Y8
		VSS_54	Y12
		VSS_55	AB3
		VSS_56	AB5
		VSS_57	AB8
		VSS_58	AB10
	A1	DNU_1	AA1
	A2	DNU_2	AA12
	A11	DNU_3	AB1
	A12	DNU_4	AB2
	B1	DNU_5	AB11
	B12	DNU_6	AB12
		DNU_7	
		DNU_8	
		DNU_9	
		DNU_10	
		DNU_11	
		DNU_12	

LPDDR4_200P

VCC_DDR
C3002 22uF X5R 6.3V C0603
C3003 10uF X5R 6.3V C0603
C3004 10uF X5R 6.3V C0603
C3005 1uF X5R 10V C0402
C3006 100nF X5R 10V C0201
C3007 100nF X5R 10V C0201
C3008 100nF X5R 10V C0201
C3009 100nF X5R 10V C0201
C3010 100nF X5R 10V C0201
C3011 100nF X5R 10V C0201
C3012 100nF X5R 10V C0201
C3013 100nF X5R 10V C0201
C3014 100nF X5R 10V C0201
C3015 100nF X5R 10V C0201
C3016 100nF X5R 10V C0201
C3017 100nF X5R 10V C0201
C3018 100nF X5R 10V C0201
C3019 100nF X5R 10V C0201
C3020 100nF X5R 10V C0201
C3021 100nF X5R 10V C0201
C3022 100nF X5R 10V C0201
C3023 100nF X5R 10V C0201

VCC0V6_DDR
C3029 22uF X5R 6.3V C0603
C3030 10uF X5R 6.3V C0603
C3031 10uF X5R 6.3V C0603
C3032 1uF X5R 10V C0402
C3033 100nF X5R 10V C0201
C3034 100nF X5R 10V C0201
C3035 100nF X5R 10V C0201
C3036 100nF X5R 10V C0201
C3037 100nF X5R 10V C0201
C3038 100nF X5R 10V C0201
C3039 100nF X5R 10V C0201
C3040 100nF X5R 10V C0201
C3041 100nF X5R 10V C0201
C3042 100nF X5R 10V C0201
C3043 100nF X5R 10V C0201
C3044 100nF X5R 10V C0201
C3045 100nF X5R 10V C0201
C3046 100nF X5R 10V C0201
C3047 100nF X5R 10V C0201
C3048 100nF X5R 10V C0201

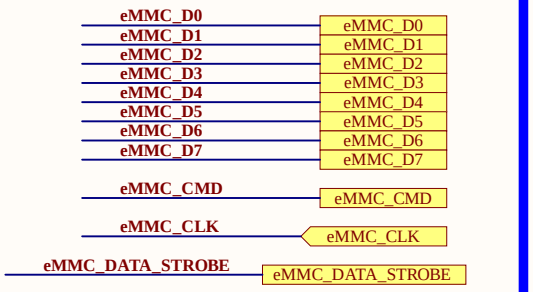
VCC3V3_SYS
C3060 4.7uF X5R 6.3V C0402
U3002 SY8089AAC SOT_23_5 (VIN, LX, GND, EN, FB/OUT)
L3001 2.2uH IND_252012 2.30A 0.102ohm
C3061 100pF C0G 50V C0402
VCC1V8_DDR
R3010 20K R0402 5%
C3066 100nF X5R 10V C0201
ETA3409A/TCS4201:1.0uH/2A
SY8089AAAC:2.2uH/2A
R3011 2K R0402 1% → VCC0V6_DDR
R3012 100K 1% R0402
Loopback from the point after sampling resistor

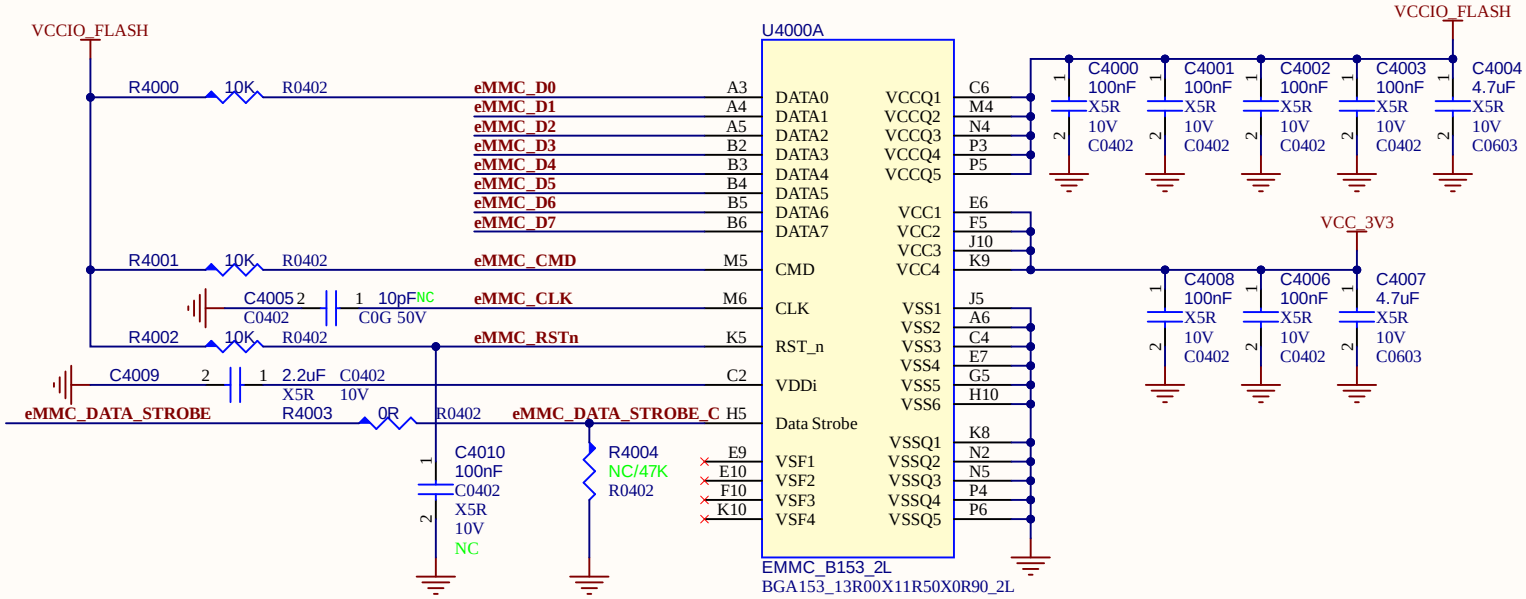
VCC_DDR
R3013 0R R0402 5%
VCC0V6_DDR_P
R3014 0R R0402 5%
C3062 10uF X5R 6.3V C0603
VCC0V6_DDR
C3064 4.7uF X5R 6.3V C0402

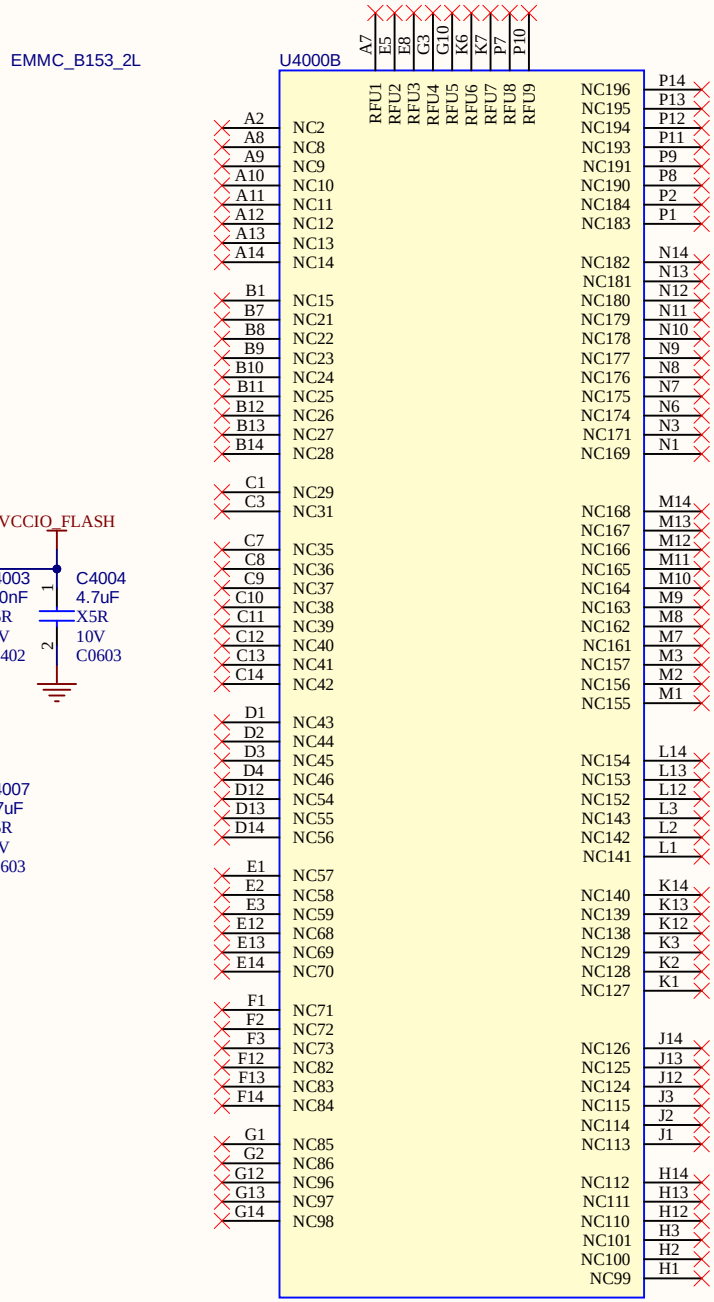
VCC0V6_DDR

VCC1V8_DDR
C3024 10uF X5R 6.3V C0603
C3025 100nF X5R 10V C0201
C3026 100nF X5R 10V C0201
C3027 100nF X5R 10V C0201
VCCA1V8_PMU
R3015 0.1R R0603 1%
C3055 4.7uF X5R 6.3V C0402
C3049 10uF X5R 6.3V C0603
C3050 100nF X5R 10V C0201
C3051 100nF X5R 10V C0201
C3052 100nF X5R 10V C0201

Flash-eMMC


eMMC_D0
eMMC_D1
eMMC_D2
eMMC_D3
eMMC_D4
eMMC_D5
eMMC_D6
eMMC_D7
eMMC_CMD
eMMC_CLK
eMMC_DATA_STROBE

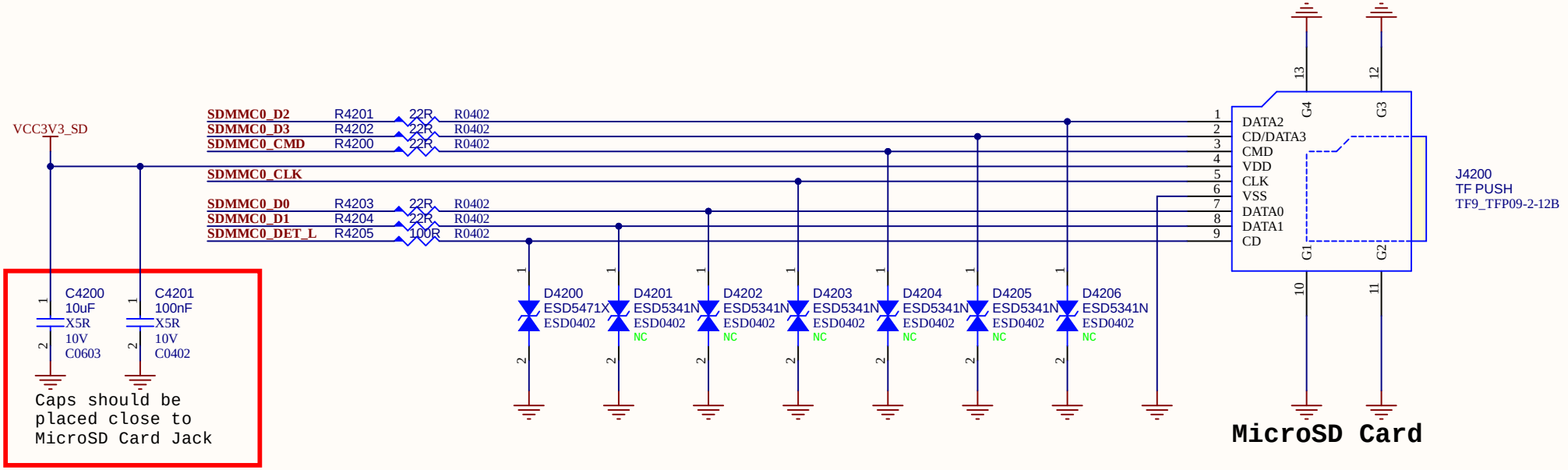

VCCIO_FLASH
R4000 10K R0402
R4001 10K R0402
R4002 10K R0402
C4005 C0402 10pF NC C0G 50V
C4009 2.2uF X5R 10V C0402
eMMC_DATA_STROBE
R4003 0R R0402
eMMC_DATA_STROBE_C
C4010 100nF C0402 X5R 10V NC
R4004 NC/47K R0402
U4000A
DATA0 DATA1 DATA2 DATA3 DATA4 DATA5 DATA6 DATA7
CMD
CLK
RST_n
VDDi
Data Strobe
VSF1 VSF2 VSF3 VSF4
VCCQ1 VCCQ2 VCCQ3 VCCQ4 VCCQ5
VCC1 VCC2 VCC3 VCC4
VSS1 VSS2 VSS3 VSS4 VSS5 VSS6
VSSQ1 VSSQ2 VSSQ3 VSSQ4 VSSQ5
EMMC_B153_2L
BGA153_13R00X11R50X0R90_2L
C4000 C4001 C4002 C4003 C4004 100nF X5R 10V C0402 C0603 4.7uF
C4006 C4007 C4008 100nF X5R 10V C0402 C0603 4.7uF
VCC_3V3


EMMC_B153_2L
U4000B
RFU1 RFU2 RFU3 RFU4 RFU5 RFU6 RFU7 RFU8 RFU9
NC2 NC8 NC9 NC10 NC11 NC12 NC13 NC14 NC15 NC21 NC22 NC23 NC24 NC25 NC26 NC27 NC28 NC29 NC31 NC35 NC36 NC37 NC38 NC39 NC40 NC41 NC42 NC43 NC44 NC45 NC46 NC54 NC55 NC56 NC57 NC58 NC59 NC68 NC69 NC70 NC71 NC72 NC73 NC82 NC83 NC84 NC85 NC86 NC96 NC97 NC98
NC196 NC195 NC194 NC193 NC191 NC190 NC184 NC183 NC182 NC181 NC180 NC179 NC178 NC177 NC176 NC175 NC174 NC171 NC169 NC168 NC167 NC166 NC165 NC164 NC163 NC162 NC161 NC157 NC156 NC155 NC154 NC153 NC152 NC143 NC142 NC141 NC140 NC139 NC138 NC129 NC128 NC127 NC126 NC125 NC124 NC115 NC114 NC113 NC112 NC111 NC110 NC101 NC100 NC99

Flash-MicroSD_Card

SDMMC0_D0 — SDMMC0_D0
SDMMC0_D1 — SDMMC0_D1
SDMMC0_D2 — SDMMC0_D2
SDMMC0_D3 — SDMMC0_D3
SDMMC0_CMD — SDMMC0_CMD
SDMMC0_CLK — SDMMC0_CLK
SDMMC0_DET_L — SDMMC0_DET_L

Note:
Take care of the type of MicroSD Card Jack,
The type here follow the logic below:
    DET is float @ card ejected; the internal pull up decided the IO pin status
    DET is connected to GND @ card inserted
If other type of Jack is used, need to shift the status of DET.



Power PMIC

I2C0_SCL_PMIC
I2C0_SDA_PMIC
PMIC_INT_L
PMIC_SLEEP_H
RESETn
PMIC_PWRON
PMIC_32KOUT_WIFI
PMIC_EXT_EN

I2S0_MCLK_M0_PMIC
I2S0_SCLK_M0_PMIC
I2S0_LRCK_M0_PMIC
I2S0_SDO0_M0_PMIC
I2S0_SDI0_M0/PDM_SDI0_M0_PMIC
PDM_CLK0_M0_PMIC

HPL_OUT
HP_SNS
HPR_OUT
SPKN_OUT
SPKP_OUT
MIC1_IN
MIC2_IN

PMIC RK809 DCDC

U2100A RK809-5A

VDD_LOGIC — Feedback from RK3562
VDD_CPU — Feedback from RK3562
VDD_GPU
VCC_DDR



BUCK1
BUCK2

DDR4	1.2V	62K 1%
LPDDR4/4x	1.1V	47K 1%
DDR3	1.53V	110K 1%
DDR3L	1.35V	82K 1%
LPDDR3	1.2V	62K 1%

Default:1.1V

PMIC RK809 LDO

U2100B RK809-5A

400mA LDO: 400mA@Vin>3V 200mA@Vin>2V

LDO1 400mA — VCC_RSV1
LDO2 400mA — VDDA_0V9
LDO3 100mA (Low noise) — VDDA0V9_PMU
LDO4 400mA (Codec vddio) — VCCIO_ACODEC
LDO5 400mA — VCCIO_SD
LDO6 400mA — VCC3V3_PMU
LDO7 400mA — VCCA_1V8
LDO8 400mA — VCCA1V8_PMU
LDO9 400mA — VCC_RSV2

2.1A SWOUT2 — VCC3V3_SD
2.1A SWOUT1 — VCC_3V3
VCC8\9: 2.7V-5.5V
BUCK5 1.5V-3.6V 2.5A — SW5 — VCC_1V8

VCC_RSV1 and VCC_RSV2 can be used for other purposes, take care of their initial voltage, it can be modified after power on.

Caps should be placed close to PMIC.

PMIC RK809 Managerment

U2100C RK809-5A

Gas Gauge

VCC_RTC — VCC3V3_SYS
BATDIV
SNSP
SNSN
EXT_EN — PMIC_EXT_EN
VDC — 1.2V
PWRON — PMIC_PWRON

I2C0_SDA_PMIC
I2C0_SCL_PMIC
PMIC_INT_L
PMIC_SLEEP_H
PMIC_32KOUT_WIFI
RESETn

VCCIO_WL
VCC3V3_PMU
VCC-5V-SYS

PMIC RK809 CODEC

U2100D RK809-5A

VCC-5V-RK

VCC_SPK_HP
CPn
CPp
VCC_CPVDD
VCC_CPVSS
VCC_1P8D (Power from VCC_RTC)
VCC_1P8A

MCLK — I2S0_MCLK_M0_PMIC
BCLK — I2S0_SCLK_M0_PMIC
LRCLK — I2S0_LRCK_M0_PMIC
SDI — I2S0_SDO0_M0_PMIC
SDO/PDMDATA — I2S0_SDI0_M0/PDM_SDI0_M0_PMIC
PDMCLK — PDM_CLK0_M0_PMIC

IO Power =LDO4

HPL_OUT — HPL_OUT
HP_SNS — HP_SNS
HPR_OUT — HPR_OUT
SPK_OUTn — SPKN_OUT
SPK_OUTp — SPKP_OUT
MIC1p (MIC_L) — MIC1_IN
MIC1n (MIC_R) — MIC2_IN

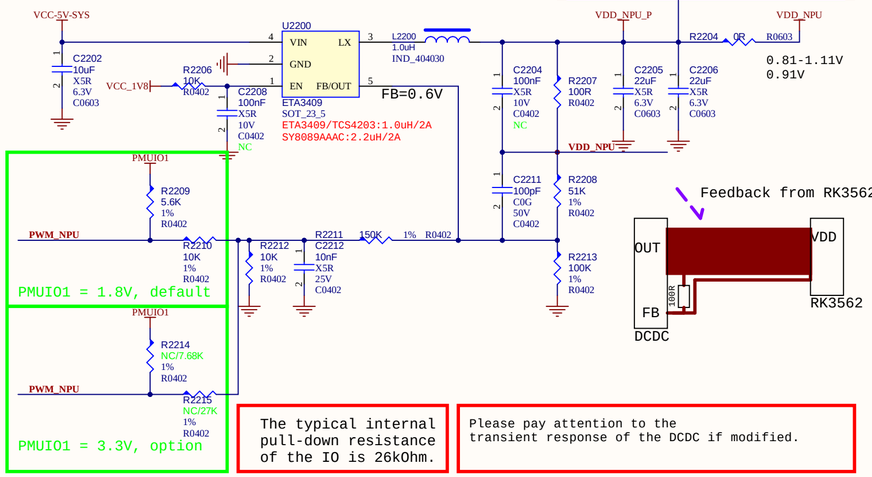
If RK809 codec is not used,
then Pin 14,15,16,17,19,40 Tie VSS
Pin 18,36,37,38,35,39,41,34,32,43,42
Leave floating

Power Extension

PWM_NPU
CAM_PWREN_H

VDD_NPU EXT(DEFAULT)

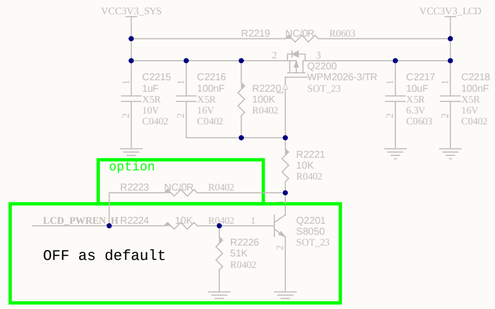
Reserve a combined power supply for costdown purposes.



The typical internal pull-down resistance of the IO is 26kOhm.

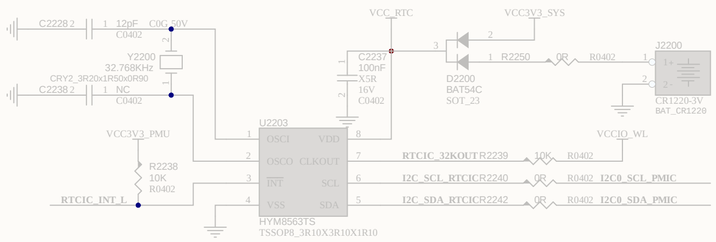
Please pay attention to the transient response of the DCDC if modified.

Power for MIPI/LVDS Panel



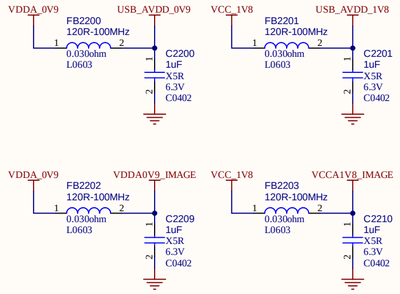
RTC IC

RTC IC do not support power-on by alarm, if function is required, plese consult RK.

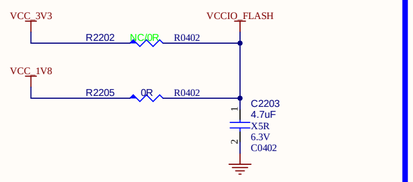


Address:Read A3H,Write A2H

Power for USB / MIPI

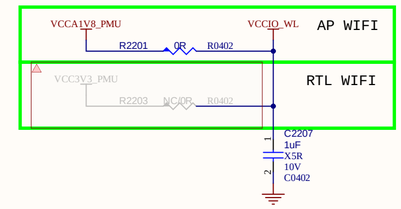


Power for Flash

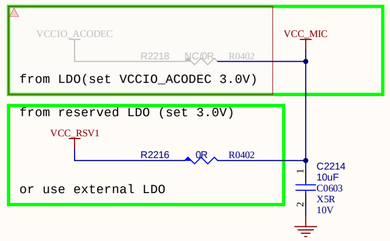


Pay attention to the voltage and match it with the actual devices.

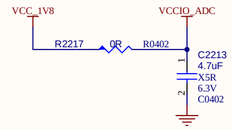
Power for VCCIO_WL



Power for MIC



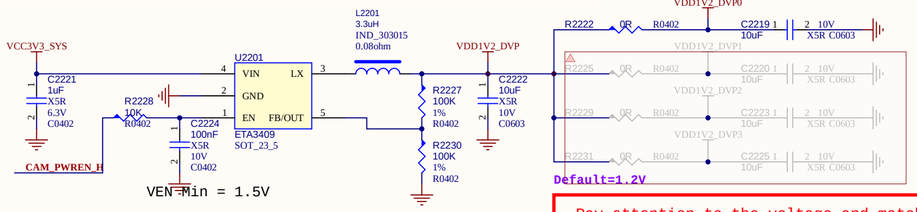
Power for Audio ADC



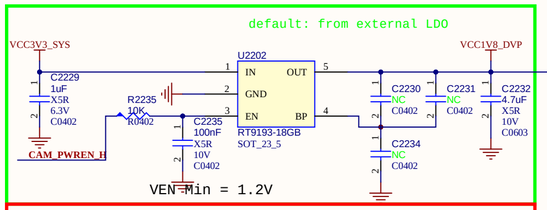
Pay attention to the voltage and match it with the actual devices.

Power for DVP

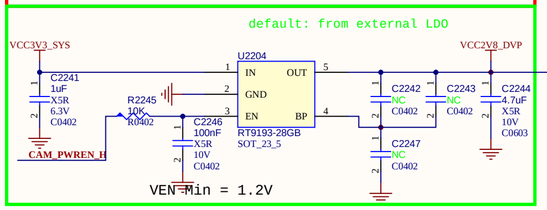
Note
Modify and adjust the circuit based on the actual situation.

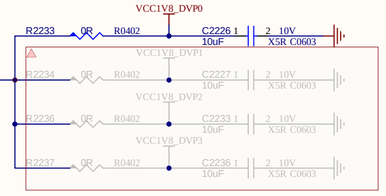


Pay attention to the voltage and match it with the actual camera module.

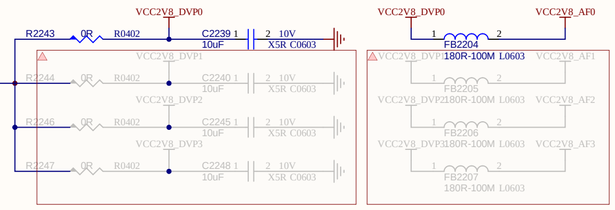


Note
VCC_RSV1 and VCC_RSV2 can be used for power supply for cameras, confirm that the initial voltage has no effect on the module, and modified them after power up.

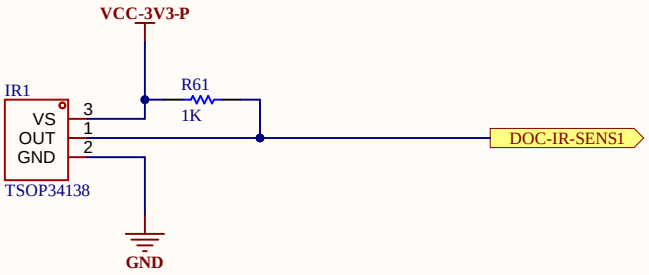


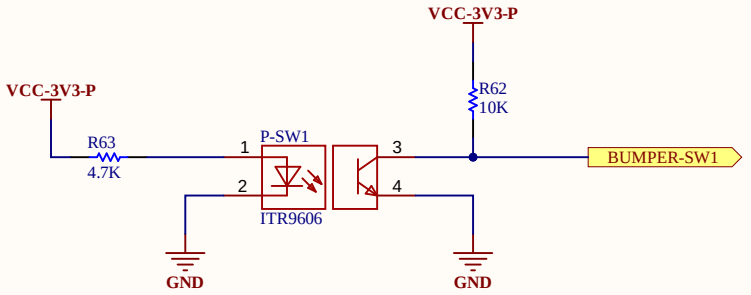


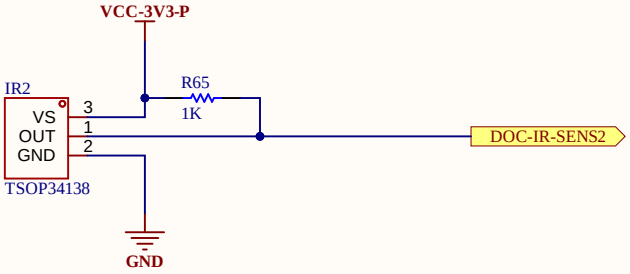
Based on the actual current, choose the path from the PMIC or an external power source.

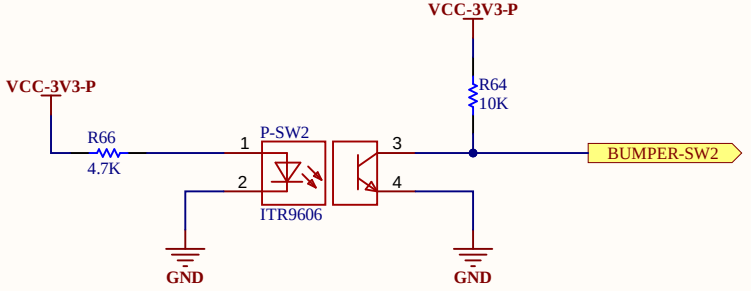


DOCKING-IR-BUMBER-SENSORs

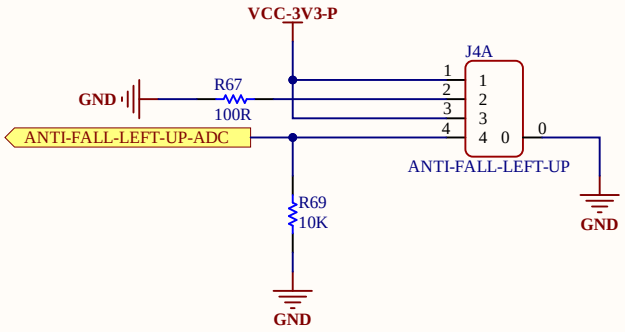


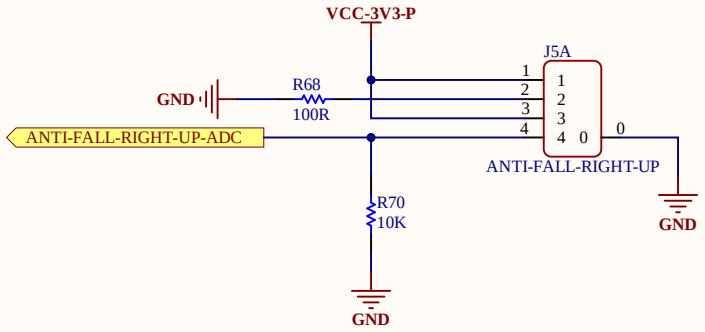


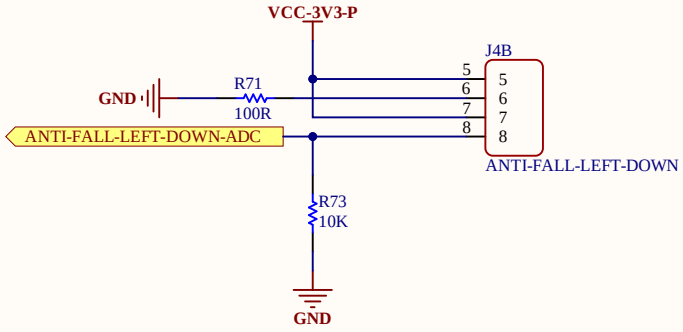


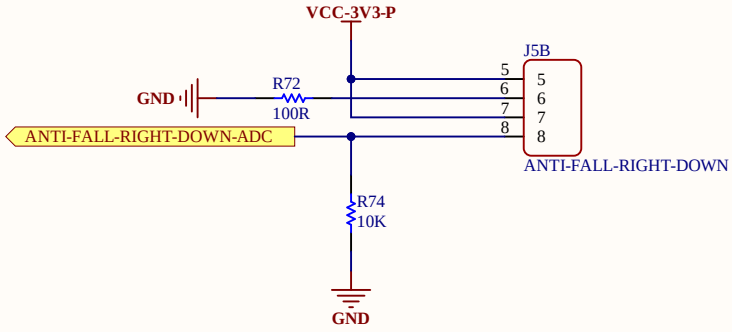


ANTI FALL SENSORs

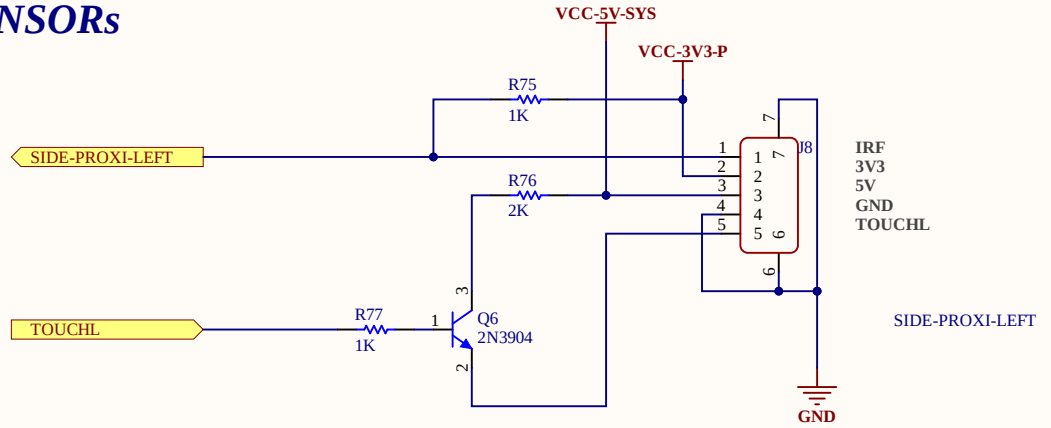


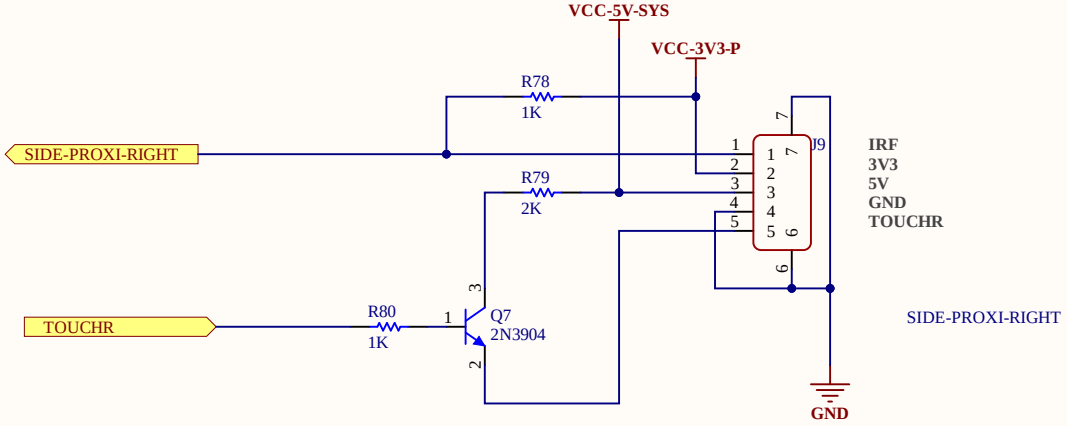




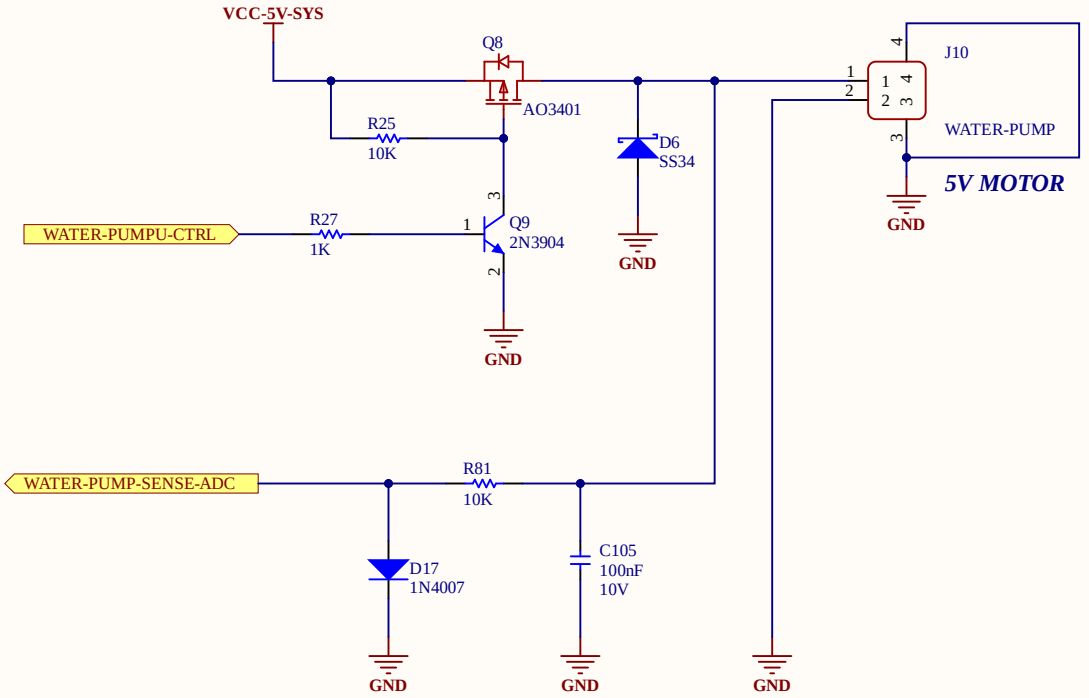


SIDE PROXI IR SENSORs

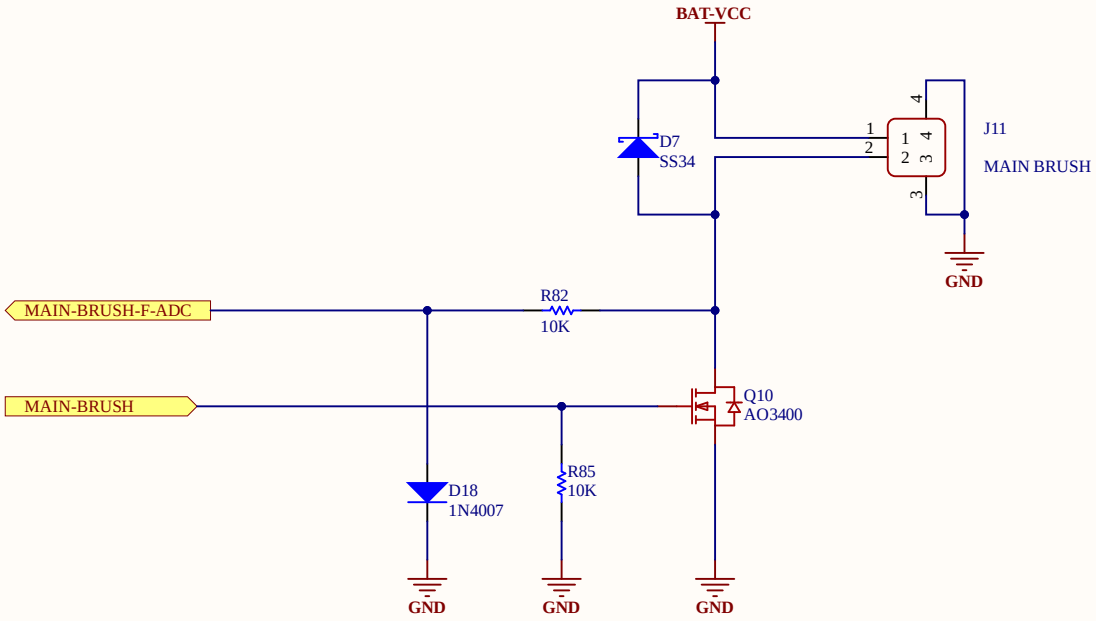




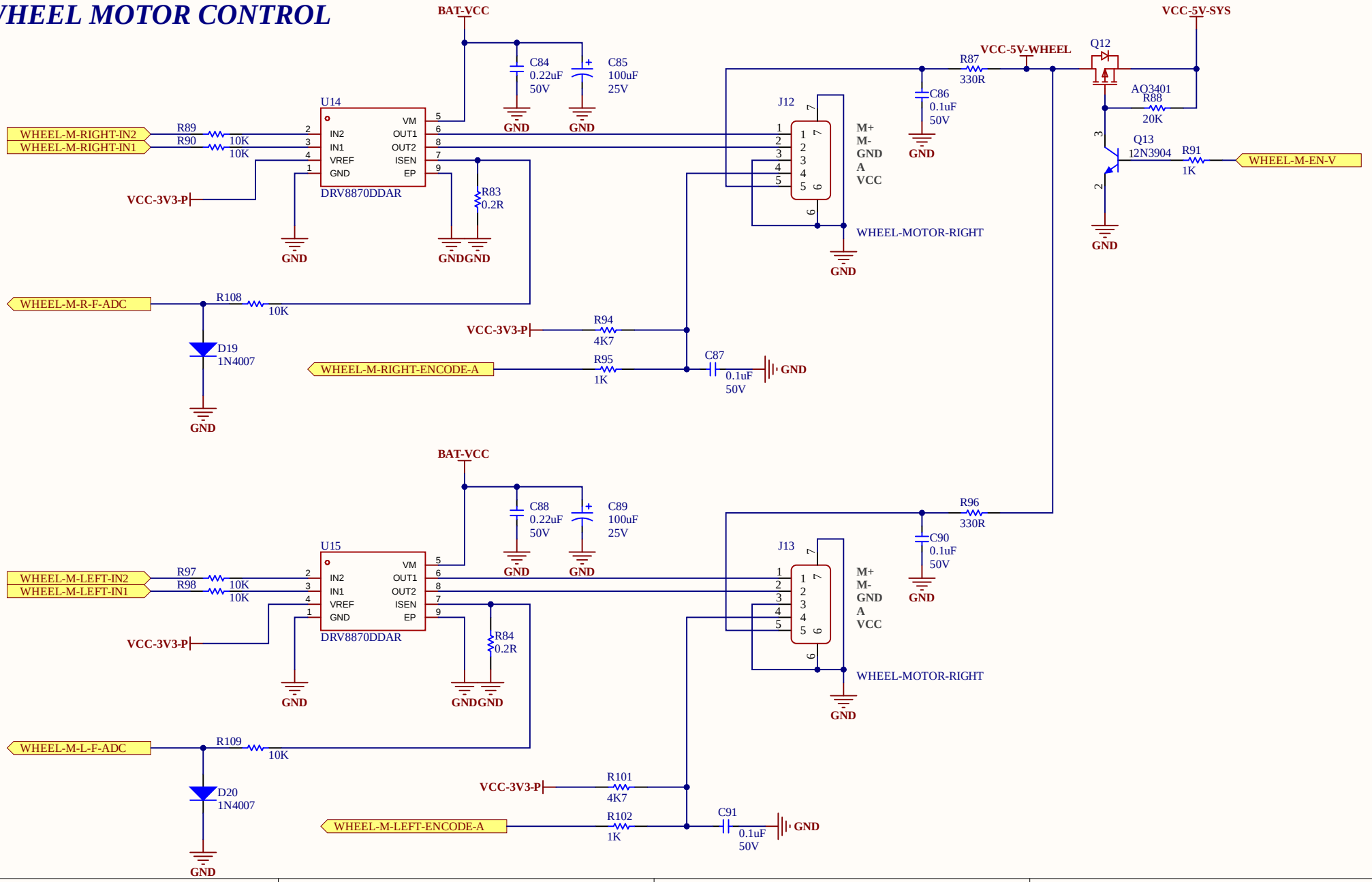
WATER PUMP MOTOR CONTROL



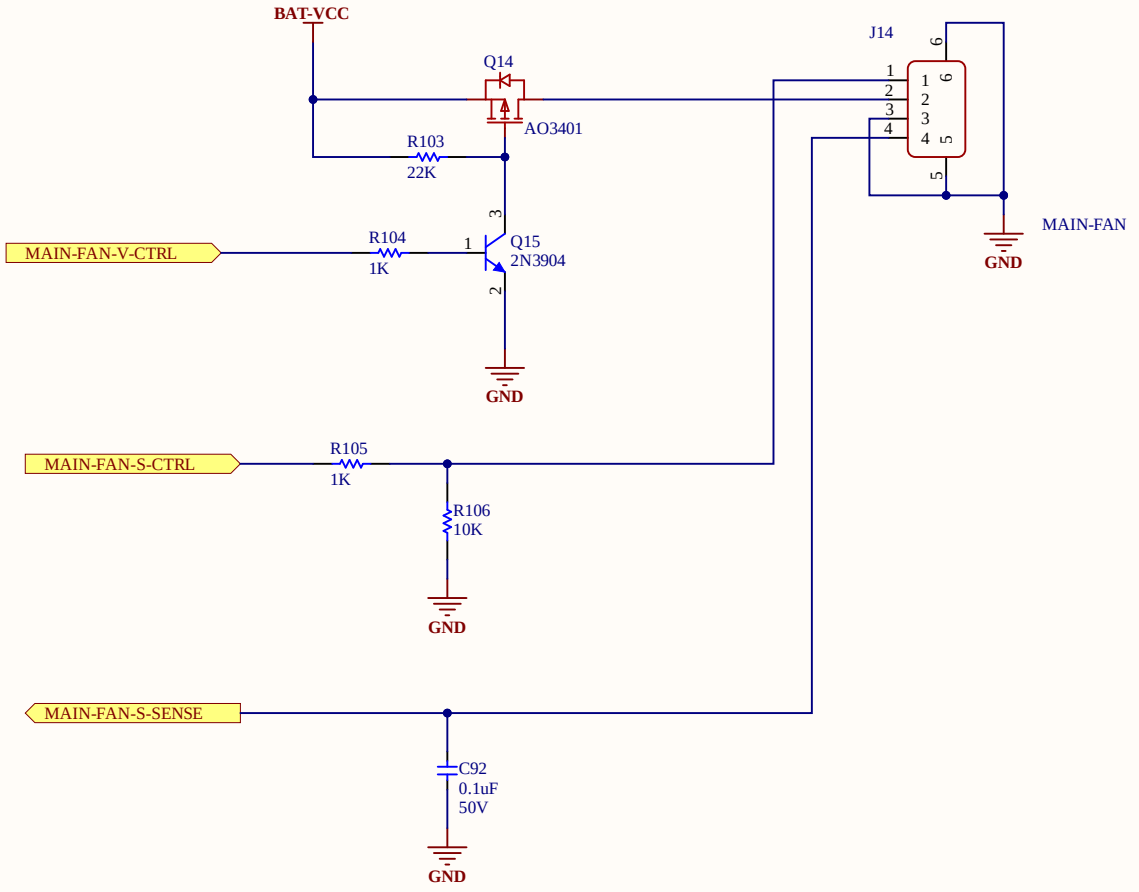
MAIN BUSH MOTOR CONTROL



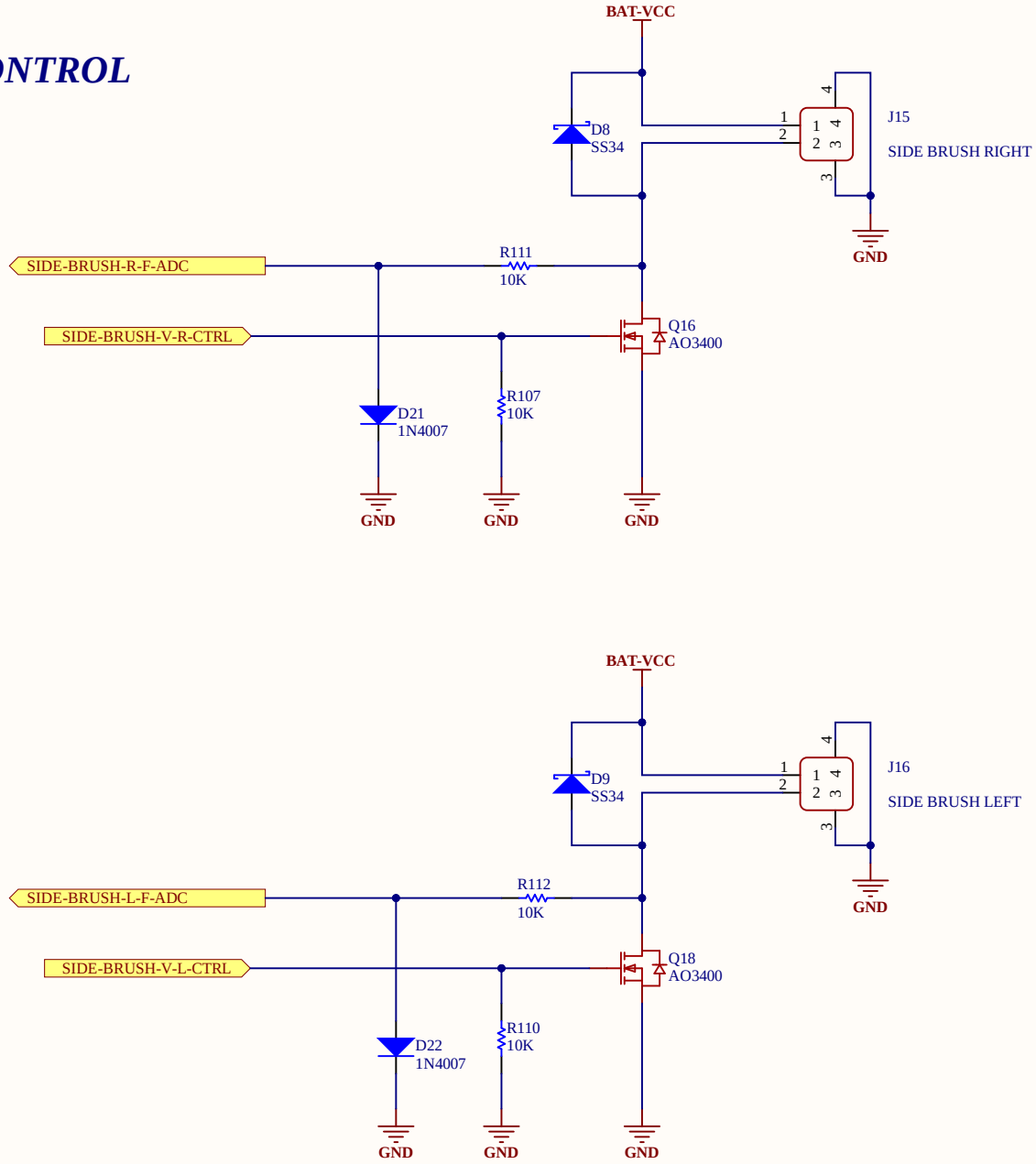
WHEEL MOTOR CONTROL



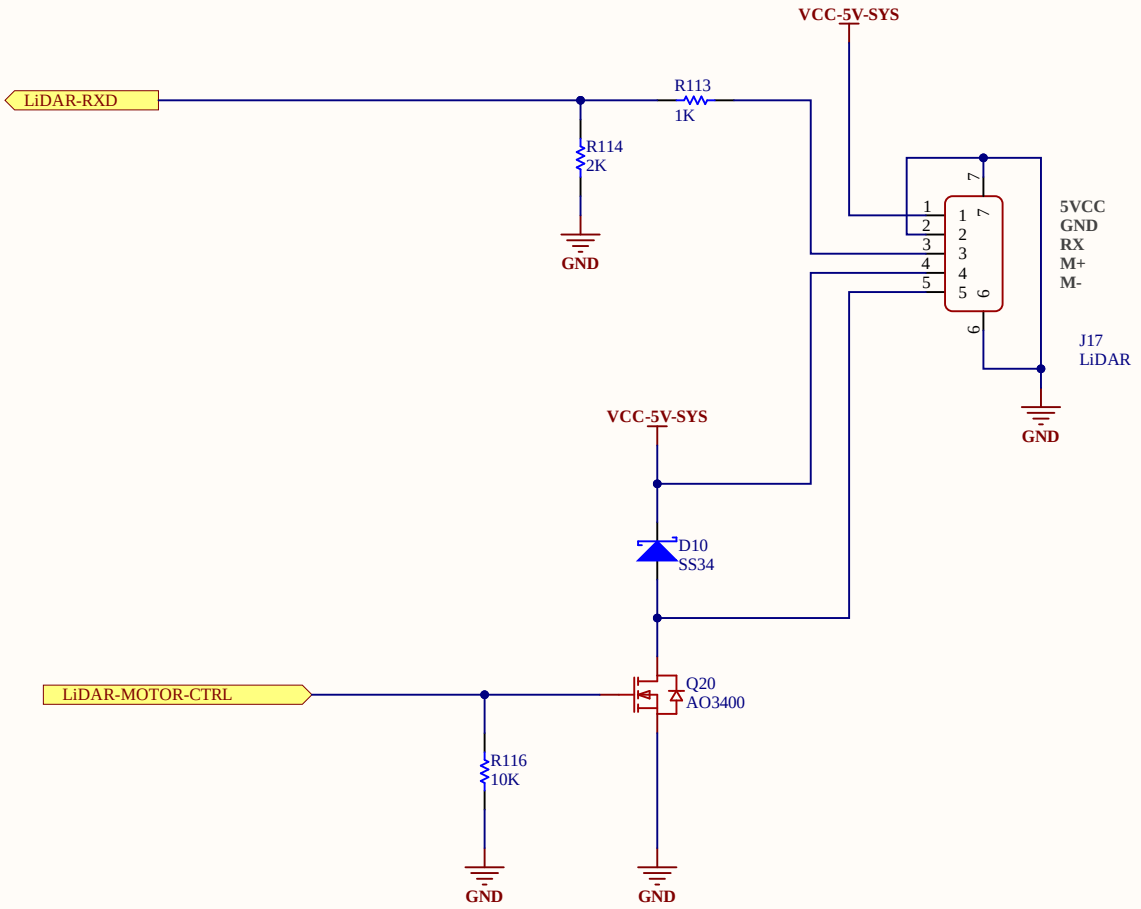
MAIN FAN CONTROL



SIDE BRUSH MOTOR CONTROL

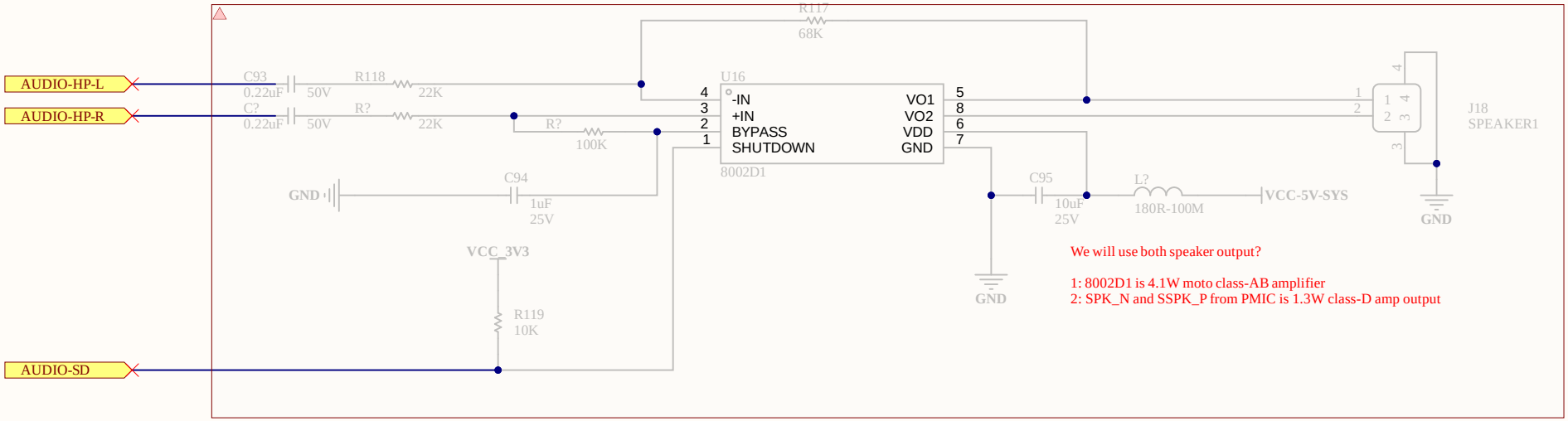


LiDAR

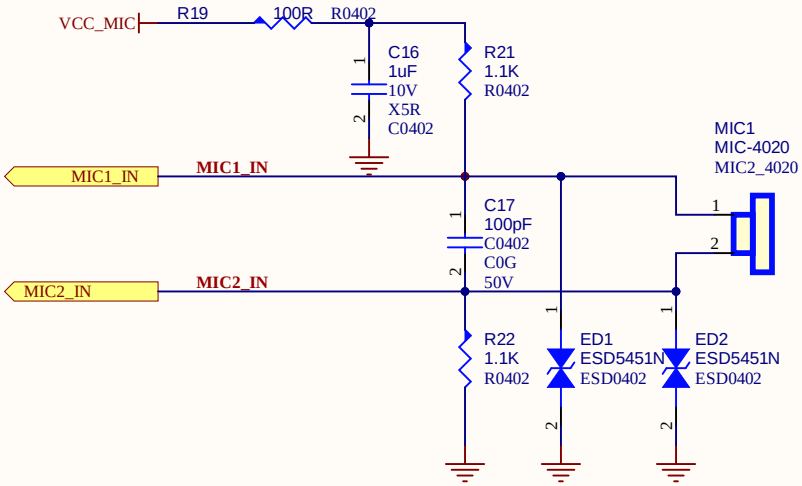

VCC-5V-SYS
LiDAR-RXD
R113
1K
R114
2K
GND
5VCC
GND
RX
M+
M-
J17
LiDAR
GND
VCC-5V-SYS
D10
SS34
LiDAR-MOTOR-CTRL
Q20
AO3400
R116
10K
GND
GND

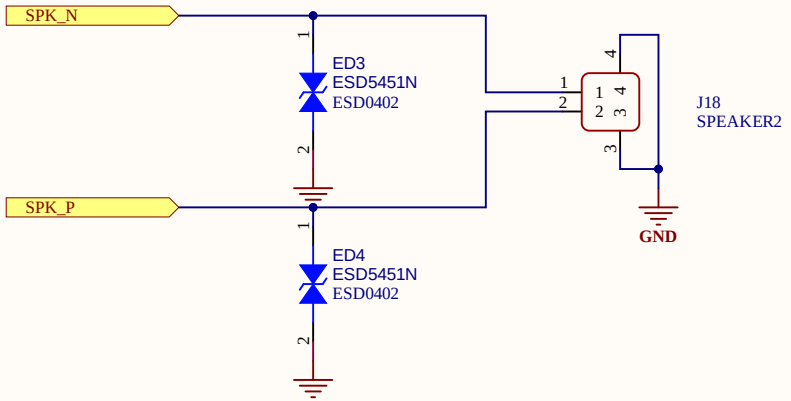
SPEAKER OUTPUT

SPEAKER MIC

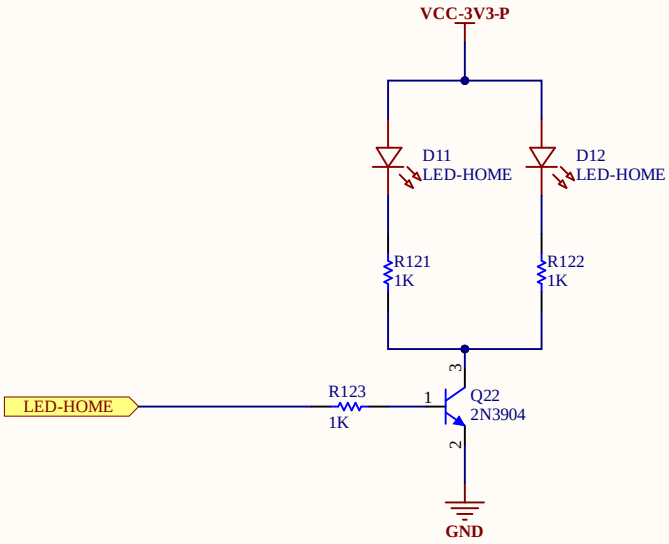


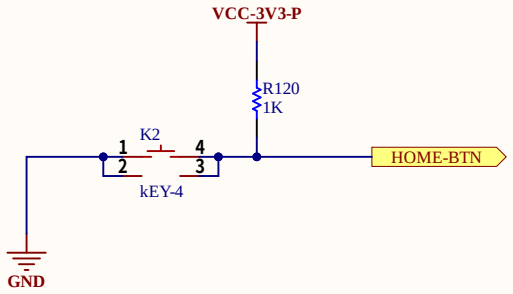
Microphone Input(Differential)

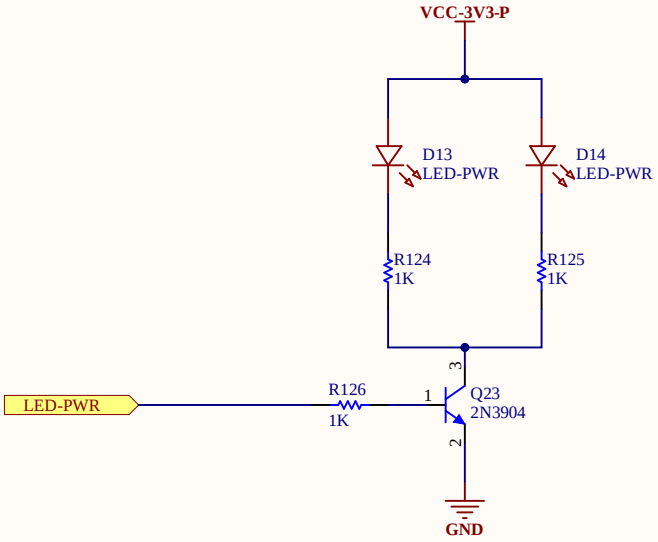


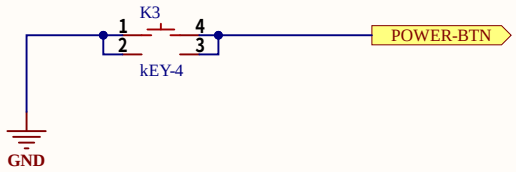


LEDs AND BUTTONs

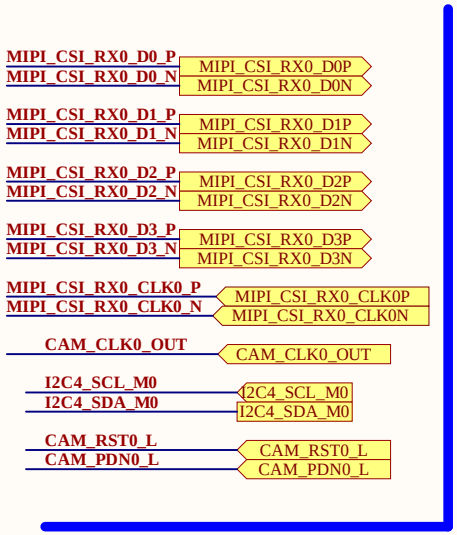


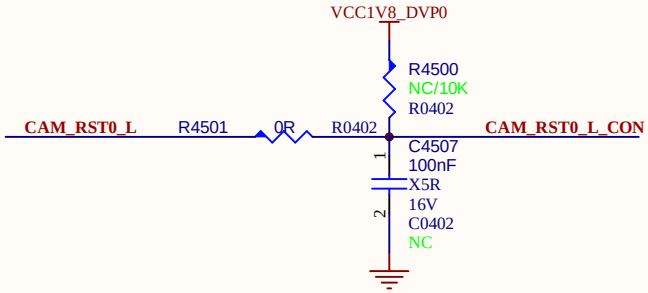


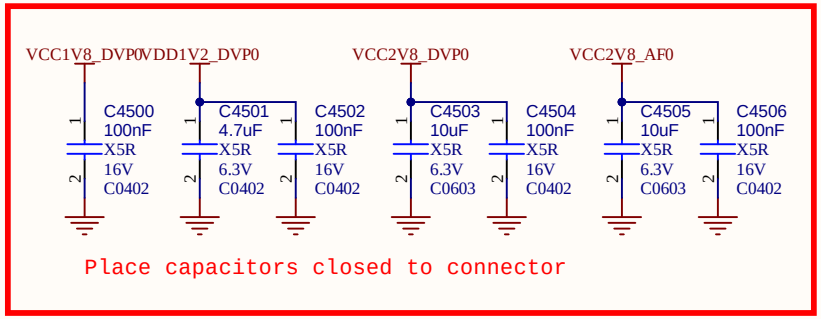


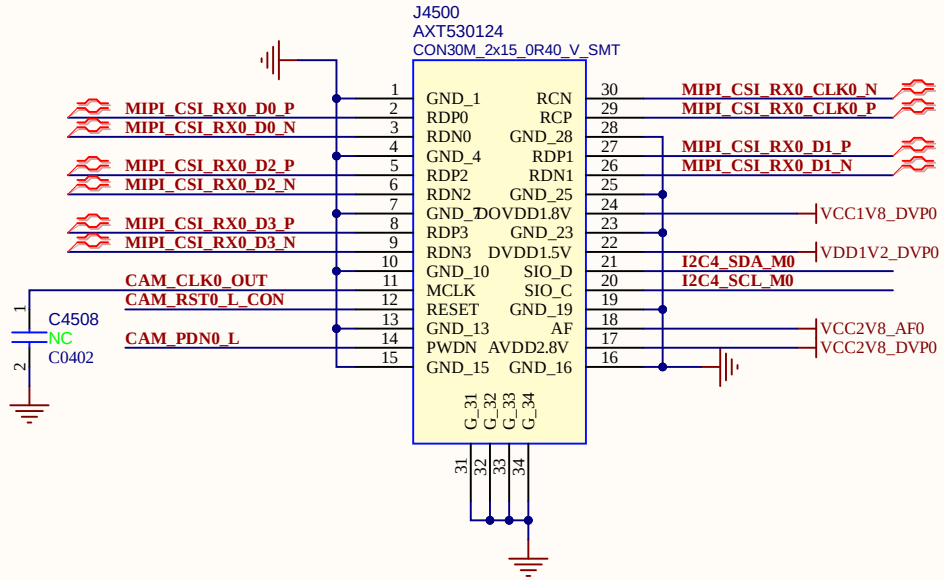


MIPI Camera
MIPI_CSI_RX0 4lanes

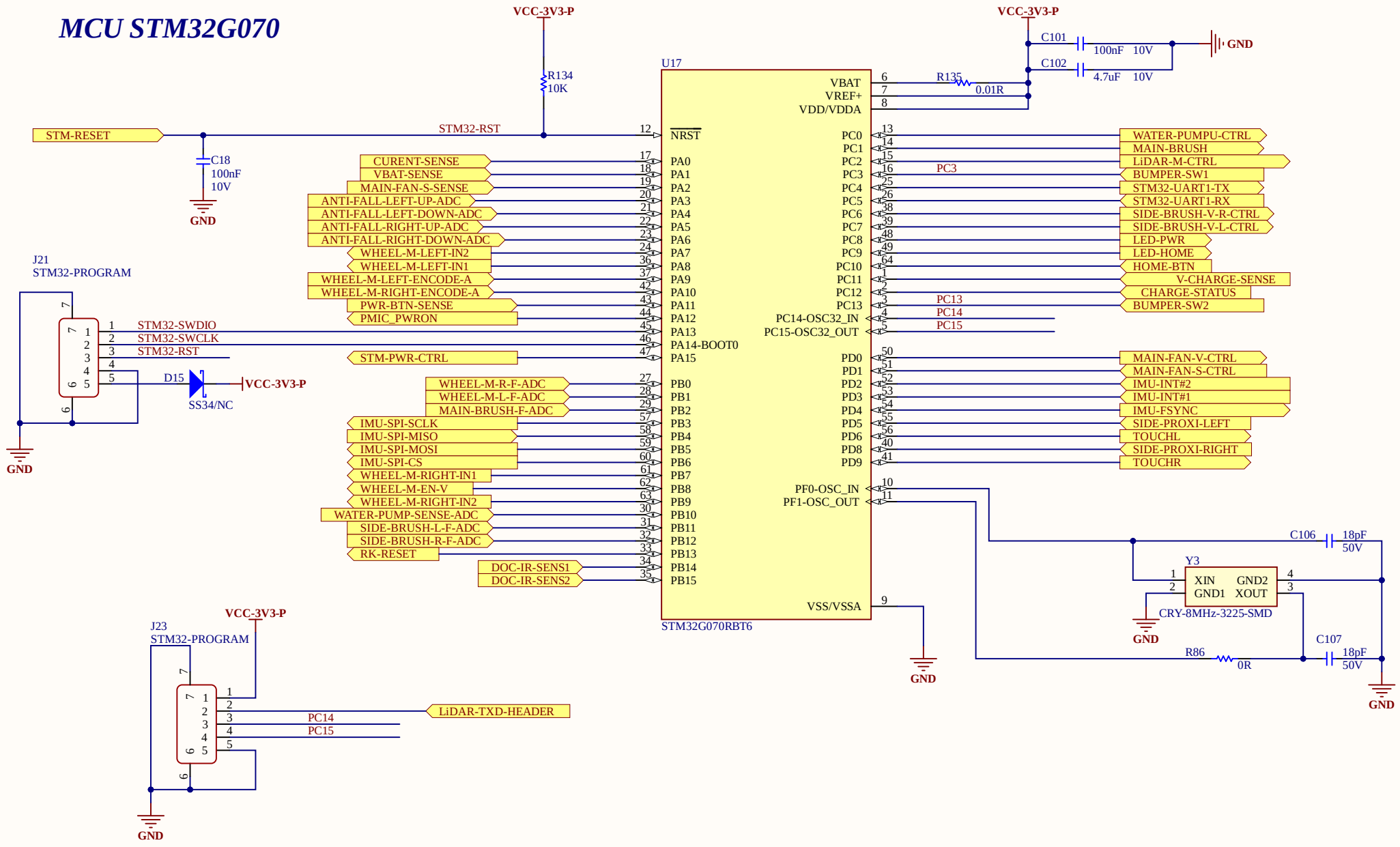




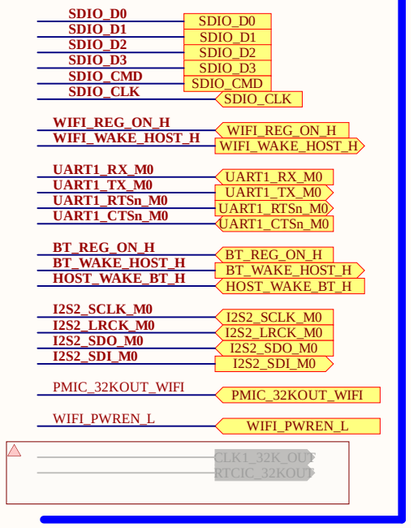


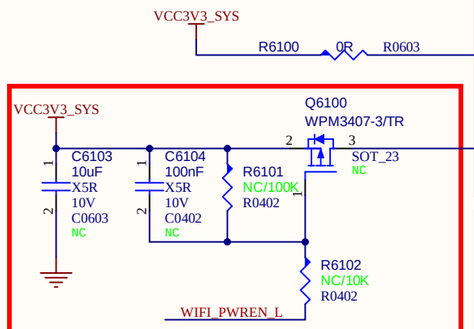


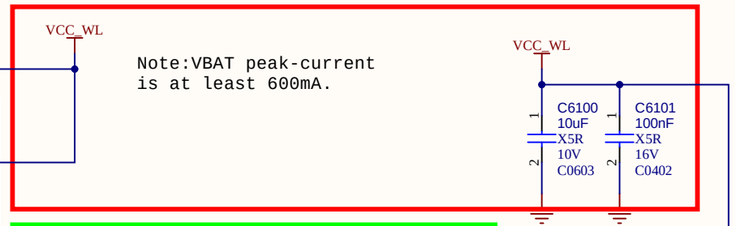
MCU STM32G070

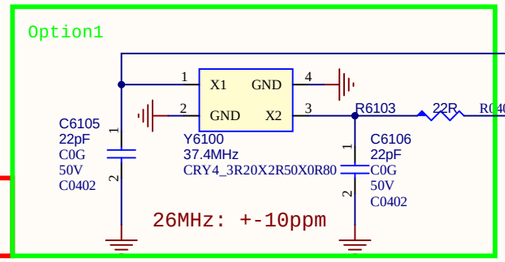


WIFI-BT-A6256

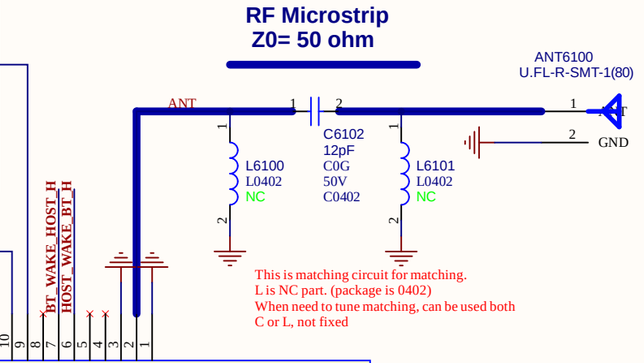


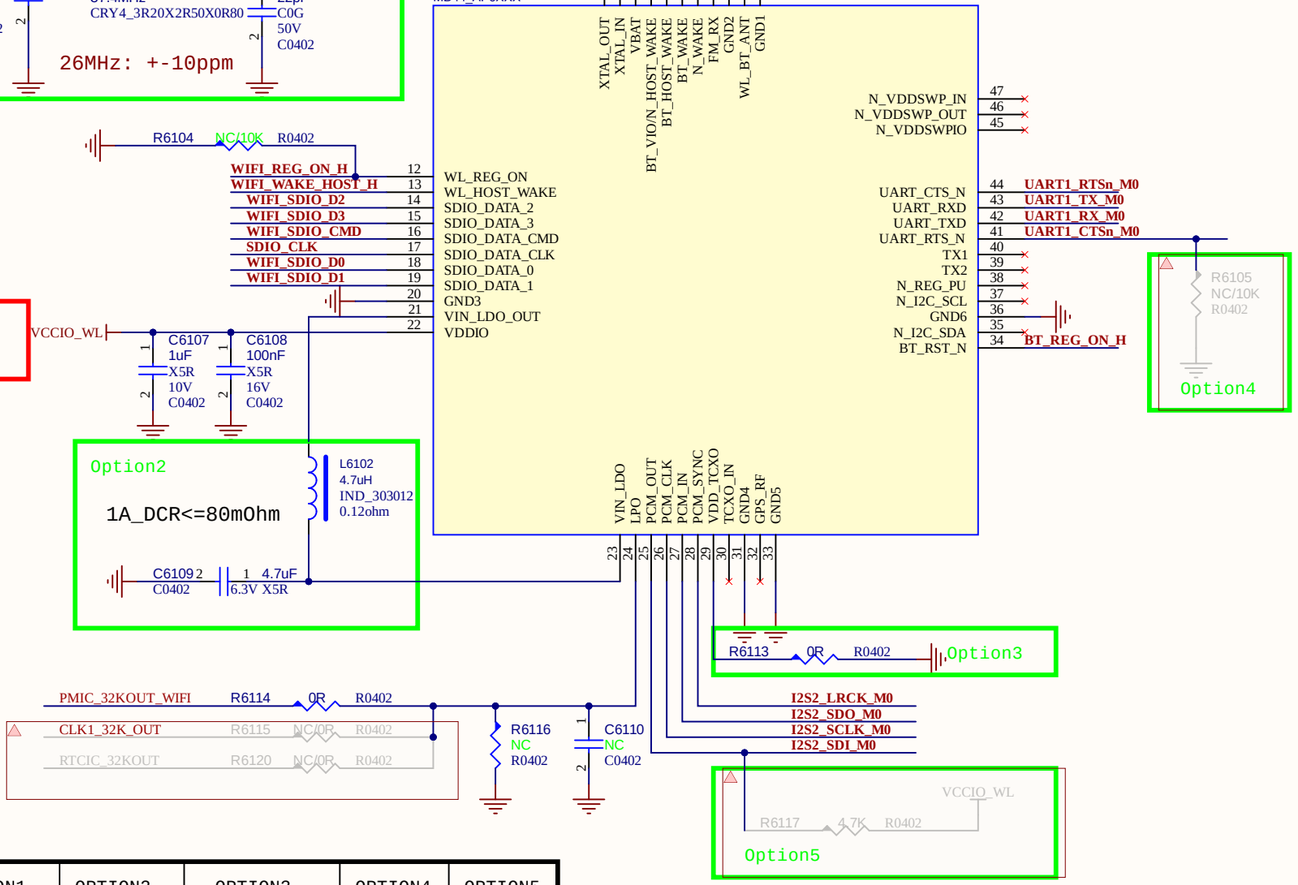




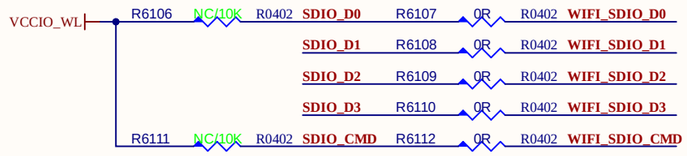


Adjust the load capacitor
according to the crystal spec.





See power page for VCCIO_WL
power connect selection.



Note:
Yes: option circuit be mounted
No: option circuit not be mounted

OPTION	WIFI				BT	Crystals	VCCIO_SDIO	OPTION1	OPTION2	OPTION3	OPTION4	OPTION5
	a	b/g/n	ac	5GHz								
AW-CM256SM	Yes	Yes	Yes	Yes	4.2	37.4MHz	1.71-3.6V	Yes	Yes	Yes@SDIO2.0 No@SDIO3.0	No	No
AP6236/AP6212	No	Yes	No	No	4.2/4.0	26MHz	1.71-3.6V	Yes	Yes	No	No	No
AP6256/AP6255	Yes	Yes	Yes	Yes	5.0/4.2	37.4MHz	1.71-3.6V	Yes	Yes	Yes@SDIO2.0 No@SDIO3.0	No	No
RTL8189ETV Module F89FTSM12-W3	No	Yes	No	No	No	Module Integrated	1.8-3.3V	No	No	No	No	No
RTL8723DS Module 6223A-SRD	No	Yes	No	No	4.2	Module Integrated	1.62-3.6V	No	No	No	Yes	Yes
RTL8821CS Module 6221A-SRC	Yes	Yes	Yes	Yes	4.2	Module Integrated	1.7-3.45V	No	No	No	No	No

IMU-ICM-42607-P

